**University of Toronto**
**Faculty of Applied Science and Engineering**
**APS111**
***Conceptual Design Specifications (CDS)***

| | | | |
|---|---|---|---|
| Team # | 173 | Date | November 23rd 2025 |

| | |
|---|---|
| Project Title | High Park Project Conceptual Design Specifications |
| Client | High Park Nature Center |
| Tutorial Section | 127 |
| Teaching Assistant | Parker Stewart |
| Communication Instructor | J. Coplen Rose |
| Prepared By<br>(Names and Student #s of Team Members) | Bryce Lin<br>Lina Habib<br>Sophie Santiago<br>Towo Oyediran<br>Adrian Kostrioukov |

3479 words State your word count (excludes Cover Page, Executive Summary, References and Appendices).

☐ Have you verified that your team has included all required sections described in the Course Assignment Reference Handbook? [Y or N]

☐ Have you included your completed and signed Attribution Table in the report submitted to Quercus AND posted a PDF of it on your MS Teams channel mentioning your TA? [Y or N]

☐ Have you used Generative AI (or Large Language Model) in the writing of this report? [Y or N] If yes, you must follow the GenAI usage and documentation course policy.

If any required components or sections are missing (including cover page or appendices ), your assignment is considered incomplete, and standard course late penalties will accrue until the submission is complete.

**Executive Summary**

The High Park Nature project seeks to design a method of transportation that balances accessibility and environmental conservation from the Grenadier Café to the Grenadier Pond. The project was commissioned by the High Park Nature Centre, a City of Toronto organization, and focuses on connecting nature with the Toronto community. Currently, the project is in its foundational phase in detailing the project requirements.

During the site visit, the paved direct pathway to the pond had steep routes ranging from 8°-18° in slope which exceeds the 15% maximum slope standard (8.5°). An unpaved pathway through the Black Oak Savannah had shown similar steepness. However, the most accessible route follows a long detour around these areas 426 metres longer than the direct route.

It was observed that most people use the paved pathway instead of the unpaved one through the Black Oak Savannah.

Given the information gathered from the client statement and site visit the design will:

- Focus on providing accessibility to people with wheelchairs and families with strollers
- Be contained within the scoped Black Oak Savannah Region

In respect to the Huron-Wendat and Petun First Nations, the Seneca and the Mississaugas of the Credit River, the Black Oak Savannah should be protected. To maintain the ecosystem, less than $250m^2$ of area should be removed. Additionally, in regards to the safety of those with accessibility needs, Canadian Standards Association and Accessibility Standards Canada require that the design abide by certain guidelines like size, slope and noise levels.

The project requirements provide the foundations for the conceptual designs. The three designs are: the Elevated Boardwalk; Birdseye Gondola; and Graded Switchback. The Elevated Boardwalk will be built in a Z-Shape with wooden signage from the cafe to the pond to fit accessibility standards. The Birdseye Gondola will be a cable car through the savannah with speakers for information and ramps following the maximum slope standard used to access it. The Graded Switchback will be a ground Z-Shape path with handrails, direction indicators and specially designed path material that increases traction and follows accessibility standards.

Considering the gap, need and project requirements, the final proposed solution is the Elevated Boardwalk since it respects accessibility standards and the Black Oak Savannah best. To proceed with the project, the client must review or select an idea. Once approval is given, construction may begin.

## 1.0 Introduction

High Park is an ecologically significant attraction located in West Toronto spanning 161 hectares. The park supports many points of interest, programs, and wildlife [1]. The client—the High Park Nature Centre—is responsible for hosting events that promote environmental stewardship. They are concerned with issues around the sustainability and accessibility of the current means of transportation around the park [2].

This document will cover the main issues and features regarding the current state of the park, review the main interest holders, and provide necessary components of a solution. The next section will cover the idea generation and selection methods employed to reach three alternate solutions. This will lead into descriptions of each design and how they achieve the required demands. The document will close by suggesting a solution that will fill the gap and need with justification of how it fulfills the client's request.

**2.0 Problem Statement**

The current gap is a lack of equitable access to reach Grenadier Pond from the café that is conservant of the environment and integrates the Black Oak Savannah.

Currently, there are two main categorizations of routes: paved pathways and unmarked trails that pass through the Black Oak Savannah.




Figure 1. Map showing paved pathways (white), accessible route (purple), and trails (red) from Grenadier Café to the pond. (Taken from [3]).

Users with mobility impairments do not have a direct route, taking an additional 426 metres to arrive at the pond compared to users without those needs.




Figure 2. Measurement of the direct route (left) and the accessible route (right). (Taken from [4]).

The incline of the direct trails to the pond exceed the national standards (8% total slope), while areas within the unpaved pathway exceed the standard of 15% [5], inhibiting access for users with mobility issues. They are limited to use the paved pathways, which do not integrate the Black Oak Savannah.

The design will implement a method of access that conserves the Black Oak Savannah while accommodating the mobility needs of its users. To “conserve” the savannah, the design will aim to maintain as much of the savannah as possible by minimizing the amount of destruction towards the existing landscape. This will be further defined in our constraints later in the document. Additionally, sustainable practices will be used to implement the design as a means of conservation. The Volunteer Stewardship Program Guidebook—a guidebook created by a volunteer group responsible for conserving, preserving and maintaining the Black Oak Savannah in High Park recognizes conservation as valid environmental stewardship [6]. This is a metric that ensures the design conserves the Savannah.

Figure 1 shows the geographical scope connecting Grenadier Café to the pond. It will consider people with wheelchairs and families with strollers, and be accessible from 7 AM until 7 PM.

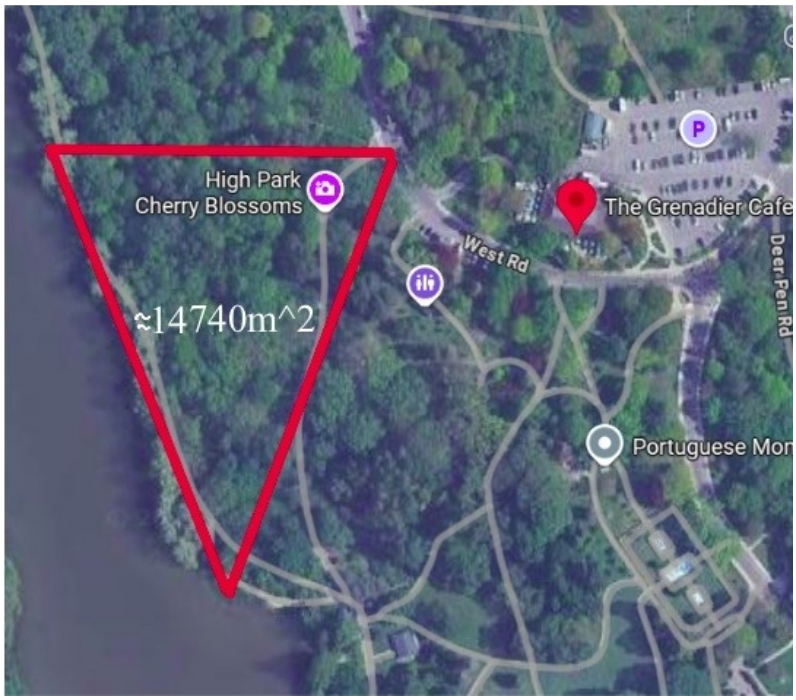


Figure 3. High Park geographical scope. (Taken from [4]).

### 3.0 Service Environment

High Park contains natural and manmade structures that interact with its users [7]. When discussing possible designs, understanding the existing environment is vital.

### 3.1 Physical Environment

There are many physical elements in High Park that impact its accessibility. Table 1 describes notable characteristics of the park environment.

Table 1. Features of the Physical Environment

| Location/Section (From Grenadier Café to Grenadier Pond) | Features of the Environment |
|---|---|
| Pathways through the Black Oak Savannah | - Tripping hazards: roots, turtle nests and rocks protruding from the ground (Appendix B)<br>- Narrow<br>- Terrain steepness ranges from 8-18° in Figure 4 (Appendix C)<br>- Elevation drop of around 15 meters down the paths shown in Figure 4 (Appendix D)<br>- Trackless Train (Hazards: disrupts pedestrians/cyclists, fast-moving vehicle) (Appendix E) |

| | |
|---|---|
| | - Temperature ranging from -10° to +25°C with record highs of about 42°C and record lows of about -25°C since 2008 [8]<br>- Precipitation ranging from 23.3 to 55.1 millimeters of rainfall with since 2008 [8] |
| Paved train pathway | - Steepness of the pathway 13° (Appendix C)<br>- People using the fencing on the pathway as a handrail (Appendix D) |

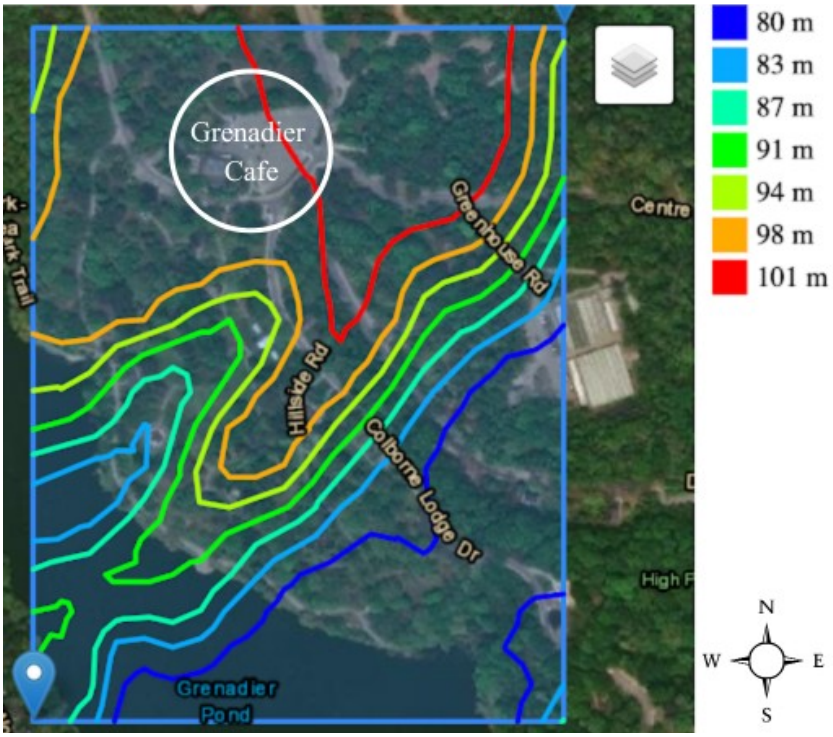


Figure 4. Elevation of High Park scope in metres (Taken from [9]).

### 3.2 Virtual Environment

The High Park area allows for visitors to use many online services. Users have access to these services as noted in Appendix F:

- Cellular data
- Phone service
- GPS

### 3.3 Living Things

High Park provides a space for various recreational activities. Groups of people can be seen engaging with the space as such: (Appendix D)

- Black Oak Savannah area

    - Used by adults without visible mobility issues
    - Fewer people present than the paved pathways
- Path by the pond
    - Age: kids-elderly
    - People jogging, strolling through the pathway
- Recreational activities
    - Exercises
    - Picnics

High Park visitors may interact with various wildlife along the paths. It is important to be aware of the following animals when journeying the park (Appendix G):

- Rabbits
- Chipmunks
- Ducks
- Bees
- Pigeons
- Red fox [10, p.18]
- Bats [10, p.20]
- Deer [10, p.25]
- Skunk [10, p.26]
- Raccoon [10, p.27]
- Coyote [10, p.28]

**4.0 Interest Holders**

When designing a solution for a populated area like High Park, the people that may have concerns over the design must be considered. The most important interest holders identified are listed in Table 2. Other brainstormed interest holders can be found in Appendix H.

Table 2. Interest Holders in order of importance

| Interest Holders | Interest and Impact |
|---|---|
| Toronto Parks and Recreation | Toronto Parks and Recreation is the chief funder and operator of High Park. Any facilities opened on their property requires their consent [11]. |
| Huron-Wendat and Petun First Nations, the Seneca, the Mississaugas of the Credit River | As the caretakers and original inhabitants of the High Park land, these Indigenous groups should be influential in the design process [12]. |
| Urban Forestry Department | The Urban Forestry Department is responsible |

| | |
|---|---|
| | for maintaining sensitive areas in High Park. Their jurisdiction includes the project's scoped area [13]. |
| High Park Natural Environment Committee | This group's main concern is to protect and preserve the endangered Black Oak Savannah [14]. |
| Disability Alliance Canada | The design will alter how the individuals with disabilities navigate the High Park environment by making the route to the pond more accessible [15]. |
| Carla Construction & Maintenance Ltd | They fund and manage the trackless train transportation inside High Park. [16] |
| High Park Stewards | Stewards that care about High Park's environment may be concerned over a design's impact on the environment [17]. |
| High Park Residence Association [18] & High Park Tenants Association [19] | Countless residents of a ~100,000 person community [20] may access High-Park to travel the neighborhood. The organizations overseeing these residents may be concerned by potential disruptions. |

## 5.0 Detailed Requirements

The design is scoped to primarily support wheelchair-users and families with strollers from the café to the pond. It will operate within the Black Oak Savannah contained in an approximately 14740$m^2$ area (Figure 1). The following functions and objectives have been chosen to support this scope.

### 5.1 Functional Requirements

This segment details primary focus on accessibility. The black-box method (Appendix I) was used to determine the core necessities to the design shown in Table 3.

Table 3. Primary and Secondary Functions

| Primary | Secondary |
|---|---|
| The design will transport people with mobility issues from the café to the pond | - Will travel horizontally<br>- Will travel vertically<br>- Will support weight<br>- Will communicate information to the user |

**5.2 Objectives**

Objectives guide a design to be successful by addressing interest holder concerns. Table 4 describes the goals that the design follows and how they help create an equitable and environmentally-conscious mode of transportation for users.

Table 4. Design Objectives

| Objective | Metric | Goal | Justification |
|---|---|---|---|
| Maintain the Black Oak Savannah | Square meters of land removed | < 250m² | Minimum width of transportation service is 1.1-1.6m. The shortest distance through the Savannah is 108m (Appendix I) [21, p.8]. Indigenous protectors of the land require that the design protects the living creatures and cultural significance of the environment [12][22]. |
| Energy efficient | Kilojoules per hour | < 50,000 kJ/h [23, pg.6] | An energy efficient system minimizes pollution inside the endangered Black Oak Savannah, maintaining High Park's pre-colonial environment as requested by High Park Nature Centre. [2, pg. 2][24][Appendix I] |
| Transport users without additional aid from others | Information stations available per 150 metres | ≥ 1 information station / 150m | A Toronto parks survey was used to find the users' preferred distances between subsequent park signage. Navigation will be easier with nearby information checkpoints to guide users. [25, p.19] |

### 5.3 Constraints

This section will describe the constraints that define the rules that the design must follow to create accessible transportation that respects the nature and groups around it. The constraints for a design are presented in Table 5.

Table 5. Design Constraints

| Constraint | Metric | Limit |
|---|---|---|
| Must be big enough to accommodate the size of wheelchairs and other traffic | Dimensions of solution in meters [21, p.3]. | Minimum width and length >1500 mm and height >2000 mm [21, p.3]. |
| Must meet the accessibility standards of ASC [5], CSA [15], TSSA [26] and AODA [27] | Degree of slope of transportation [5]. | Must not exceed a slope of 15% [5]. |
| Must maintain a noise emission limit provided by OME for Class 1 Areas | Noise emission in dBA; decibels using A-weighted scale [28]. | Must not exceed 50 dBA during system operating hours; 7:00-19:00. [28, section B7.1] |

## 6.0 Alternative Design Generation, Selection and Description

Our ideas have been generated in consideration of our scope, functions, objectives and constraints. It focuses on improving transportation of mobile aids on the steep slopes between the Grenadier Café and Pond. The favoured designs valued individual independence, minimized negative impacts on the environment, and met the Canadian accessibility standards outlined in our constraints.

### 6.1 Idea Generation

Our team used four idea generation methods shown below in Figure 5. These methods helped us produce 75 potential solutions within the design space.

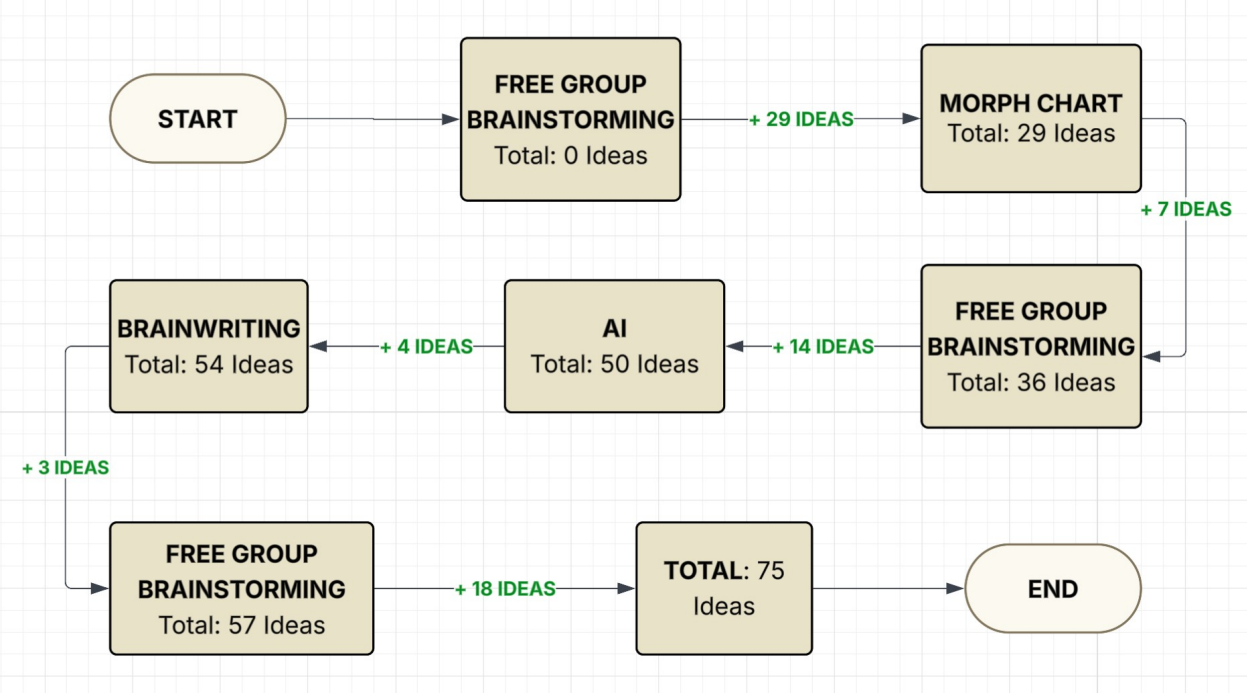


Figure 5. Flow chart showcasing our process of idea generation (Taken from [29])

Steps:

1. Brainstorming our initial ideas for 30 minutes, generating 29 ideas (Appendix K).

2. Developing a morph chart using secondary functions to generate ideas related to communication and ground, water, and air travel movement (Appendix L).

3. Brainstorming again, expanding on different methods of functionality (Appendix K).

4. Consultation of AI programs such as Copilot and ChatGPT to discover unique solutions (Appendix A).

5. Brainwriting method to develop three more ideas (Appendix M).

6. Group brainstorming session for abstract suggestions, generating 18 ideas (Appendix K)

All of these ideas were then sorted into 15 categories based on their core components or mode of transportation (Appendix N).

### 6.2 Alternate Design Selection

After finalizing our ideas, they were narrowed down by following the four step process in Figure 6.

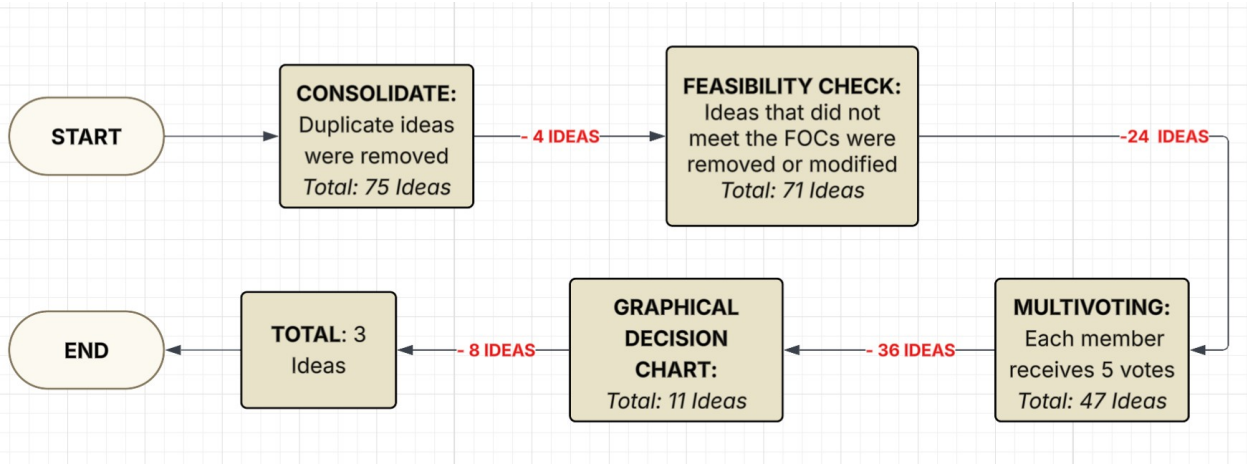



Figure 6. Flow chart of our idea finalizing process (Taken from [29])

Steps:

1. Numbered all generated ideas to organize them (Appendix N). Removed duplicate ideas to reduce the list.

2. Eliminated ideas that could not be modified to fit the requirements of our functions and constraints (Appendix N)

3. Used the multivoting process to create a reduced and realistic list of ideas (Appendix O).

4. Imputed remaining ideas into a graphical design chart to compare them against each other and our main objectives (Appendix P).

Our graphical decision chart narrowed our decision to five main ideas. Two of these were path-based ideas, so we combined those two along with the Z-Shaped Pathway to better meet the objectives. In addition, we chose the elevated boardwalk combined with the Z-Shaped path and the gondola due to their ease of use and transportation capabilities.

### 6.3 Alternate Conceptual Designs

Our team proposes three designs:

1. Elevated Boardwalk
2. Birdseye Gondula
3. Graded Switchback

#### 6.3.1 Elevated Boardwalk (Design 1)

The first design transports users using an elevated pathway. It prioritizes maintaining the Black Oak Savannah and achieves the function through its three components:

1. Elevated Platforms
2. Z-Shape
3. Wooden Signage

The design is a two-way z-shaped path that is approximately 4.0m wide and spans 163.4m that transports mass horizontally and vertically. The path is wide enough to accommodate two wheelchairs and transport other users with mobility issues. The z-shape increases the length of the route, introducing a more gradual decline down the slope to meet the slope constraints. Wooden signage will communicate information to the users. See Figure 7 for a conceptual overview of the path and its dimensions.



Figure 7. Conceptual Overview of Boardwalk Route [4].

The elevated boardwalk will require at least 132 posts placed 2.5 m apart [30] along both sides of the boardwalk with cross-bracing between adjacent posts for load distribution and safety [31]. With a diameter estimate of about 0.305 m, the approximate land area removed will be $132(\pi(0.1525)^2) = 9.64m^2$ which is well within our objective even with a larger post size.

See Figure 8 for the conceptual design layout and Appendix R for a detailed description of the boardwalk design's features and materials.

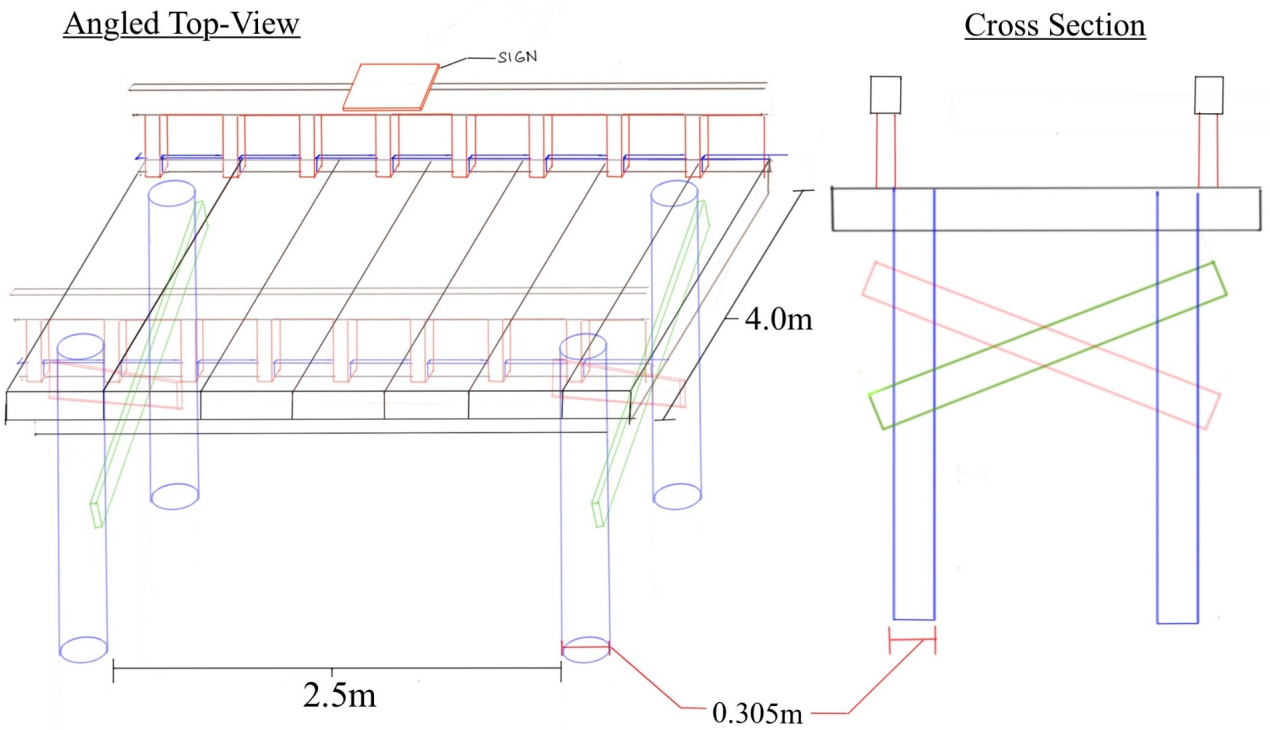



Figure 8. Conceptual Boardwalk Design

Table 6. Elevated Boardwalk Objectives Justification

| Objectives | Justification |
|---|---|
| Maintain the Black Oak Savannah | Elevated design allows for minimal interference with the Black Oak Savannah.<br>- Estimated land removal is 3.86% of the goal area. |
| Energy Efficient | No energy required for a path-based solution. |
| Transport without the aid of others | The boardwalk utilizes wooden signage to indicate direction of travel and approximate distance to the destination. Other important warnings may be added. |

### 6.3.2 Birdseye Gondola (Design 2)

The second design transports users with a cable car and pulley mechanism. It prioritizes accessibility and ease-of-use for wheelchair users. The main features are:

1. A cable car

    a. Handrails inside the cable car
2. An accessible ramp
3. Audio speakers

This design is a 2.00m x 1.50m x 2.00m cable car (Appendix R) which transports mass vertically and horizontally from the Grenadier Cafe to the Grenadier Pond. The cable car supports the mass of users with mobility issues, with a large enough area to accommodate wheelchair users as well as other individuals with mobility issues (Appendix R). Audio speakers will be used to communicate instructions and directions for individuals using the design. Handrails may be used for support as the cable car travels through the air. Additionally, a 1.0m x 0.5m ramp will provide an accessible entrance for wheelchair users. See Figure 9 for a conceptual overview of the route.

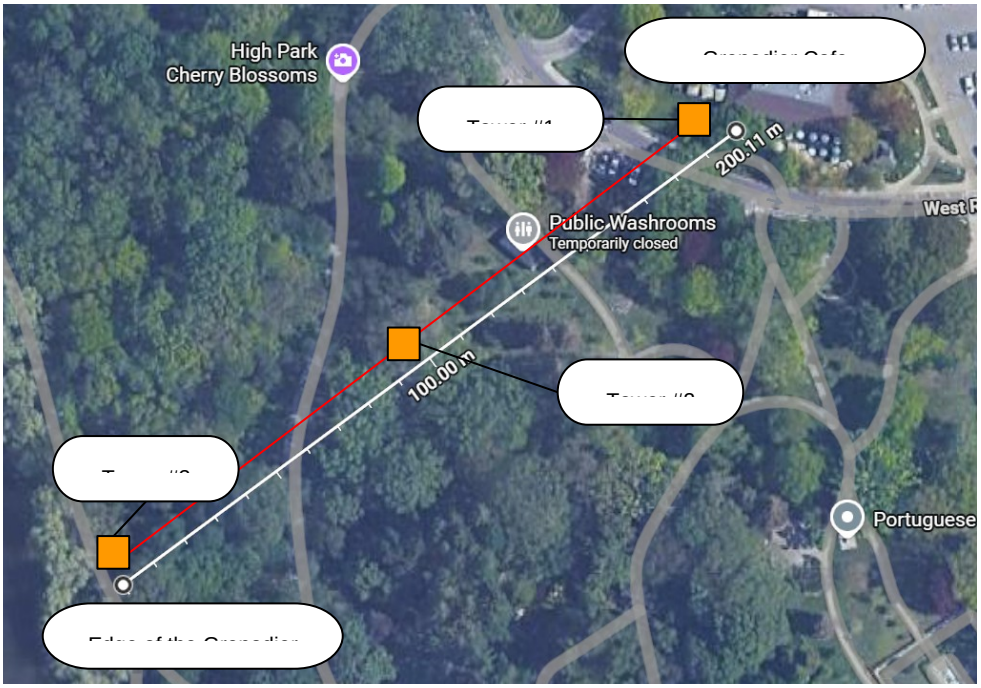


Figure 9. Location of gondola lift towers and the planned path of travel [4].

See Figures 10 and 11 for a conceptual design layout of the cabins and towers.

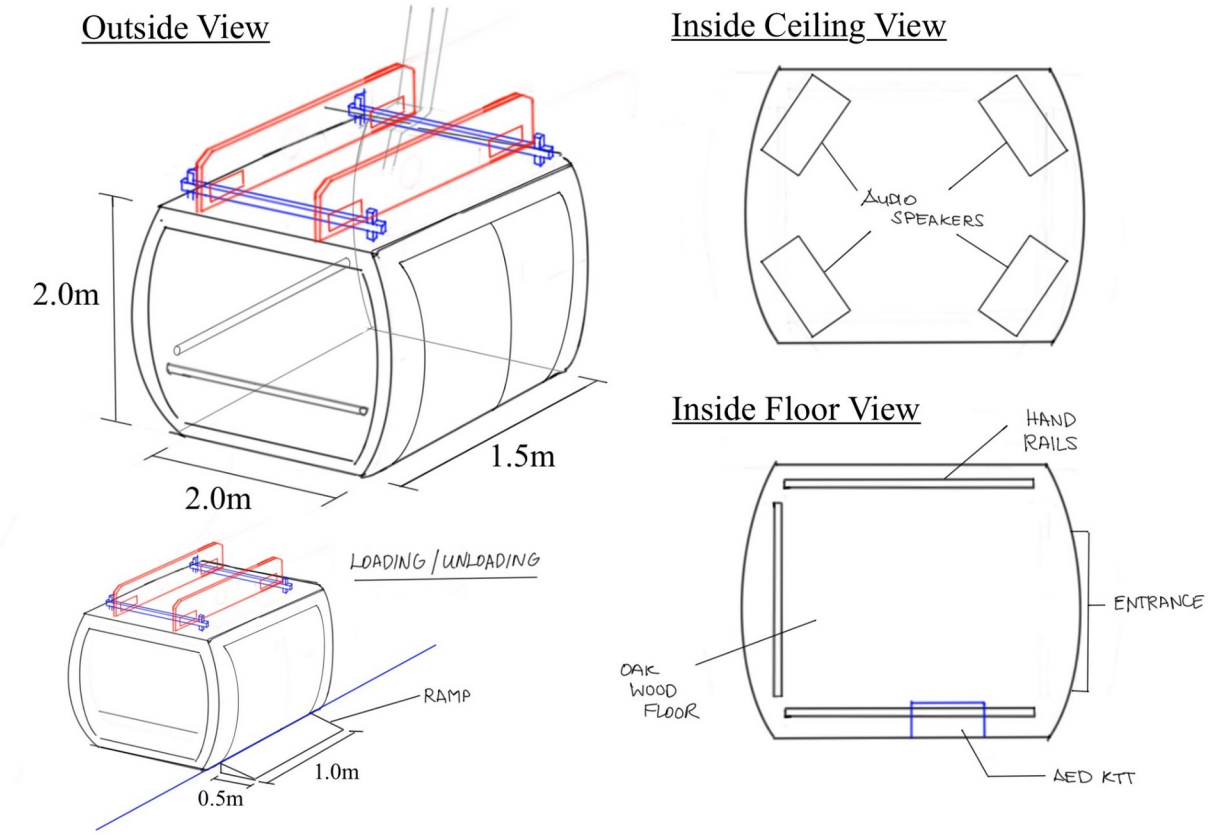



Figure 10. Theoretical gondola cabin design

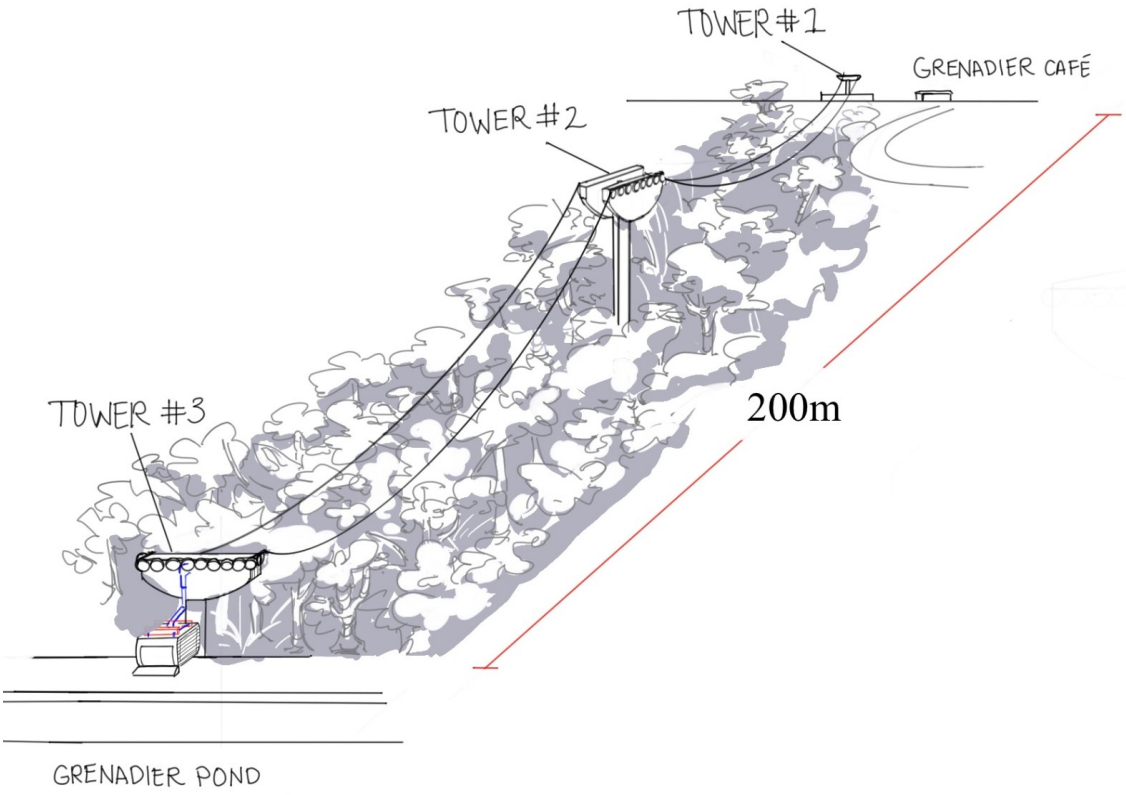



Figure 11. Conceptual design of the gondola system between Grenadier Cafe and Grenadier Pond

Table 7. Gondola Objectives Justification

| Objectives | Justification |
|---|---|
| Maintain the Black Oak Savannah | The gondola lift will have 3 towers, minimizing the destruction of the Black Oak Savanna due to the limited area removed. See Appendix R for dimensions. |
| Energy Efficient | The gondola lift draws electricity from the Ontario electricity grid, which is 50.9% nuclear energy, minimizing pollution to the environment. [32] |
| Transport without the aid of others | Audio speakers will convey important information and directions to users. |

### 6.3.3 Graded Switchback (Design 3)

The third design transports users with a traditional trail through the Savannah. It prioritizes energy efficiency and ease of travel for users on the ground. It is composed of three main components:

1. Z-Shape Path
2. Handrails
3. Metal Signage

This friction-based pathway is 3.0m wide, sufficient for two wheelchairs, and spans 163.4m that transports mass horizontally and vertically. The z-shape provides the appropriate length for a more gradual slope within the 15% constraint than a direct route. The material will consist of resin-bound gravel to increase slip resistance especially in wet conditions [33]. The handrails ensure added safety for all users and metal signage will communicate all necessary information to users. The switchback will follow the same route as the proposed design: Elevated Boardwalk (Figure 7). See Figure 12 for the conceptual design of the path and its dimensions and Appendix R for a detailed design description.

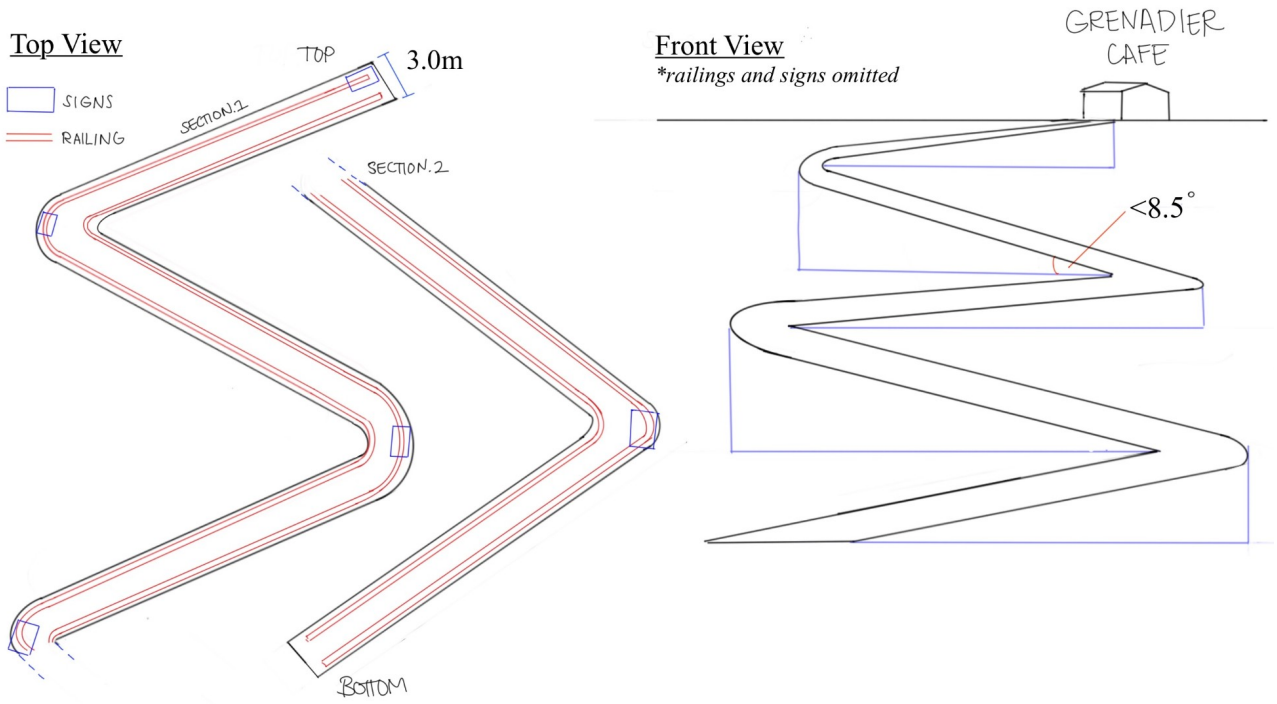



Figure 12. Conceptual Design of Graded Switchback

Table 8. Switchback Objectives Justification

| Objectives | Justification |
|---|---|
| Maintain the Black Oak Savannah | This design modifies existing paths into the z-shape to meet the slope requirement and maximize conservation of Black Oak Savannah. Width is just sufficient enough for the required constraints. |
| Energy Efficient | No energy required for a path-based solution. |
| Transport without the aid of others | The metal signage communicates directions to users and warnings to users. |

## 7.0 Proposed Conceptual Design

The Pugh Chart in Appendix S was utilized to reach the final proposed conceptual design. A 5-point grading system was used to score the efficacy of each alternative design at reaching each of the objectives of the design.

Through a careful analysis of the client's statement, detailed requirements, and the Pugh Chart, the team proposes the Elevated Boardwalk as the solution. It achieves the client's request for an equitable mode of transportation while providing a route immersing users through the Black Oak Savannah. The elevated platforms of the design removes tripping hazards that made the original pathway difficult to traverse. This design reduces travel distance by nearly 500 meters, creating a pathway that encompasses the Black Oak Savannah while being more accessible for users with mobility issues (Figure 13).




Figure 13: Side-by-Side Comparison of a) the existing pathway and b) the proposed design [4].

The choice of the elevated path involves certain tradeoffs. The installation of the elevated boardwalk would require the installing 132 load-bearing posts into the Black Oak Savannah.

Comparatively, the gondola requires 3 towers to operate, requiring the least amount of land removal for installation but the largest amount of energy to operate [Appendix Q][Appendix R]. This increases the maintenance costs over time due to added personnel needed to maintain all components of the design [34]. Thus, the elevated boardwalk is an advantageous design compared to the Birdseye Gondola since it requires no energy to run as per our objectives, lowering maintenance costs with a relatively small compromise in land removal that is well-within the objective [Appendix R]. This achieves the clients personal goals outlined in the client statement of being a low-cost design that maintains the Black Oak Savannah.

The Graded Switchback requires no energy to operate fulfilling the energy-related objective. However, the resin-bound gravel disrupts the Black Oak Savannah far more than the elevator boardwalk would as it would completely upheave the current pathway in order to create the design. The graded switchback pathway is 164 m long and 3 m wide, requiring 492 $m^2$ of Black Oak Savannah to be completely renovated. Hence, the Elevated Boardwalk is the favoured design as it minimizes the disruption of the environment that the Graded Switchback causes while being more energy-efficient than the Birdseye Gondola.

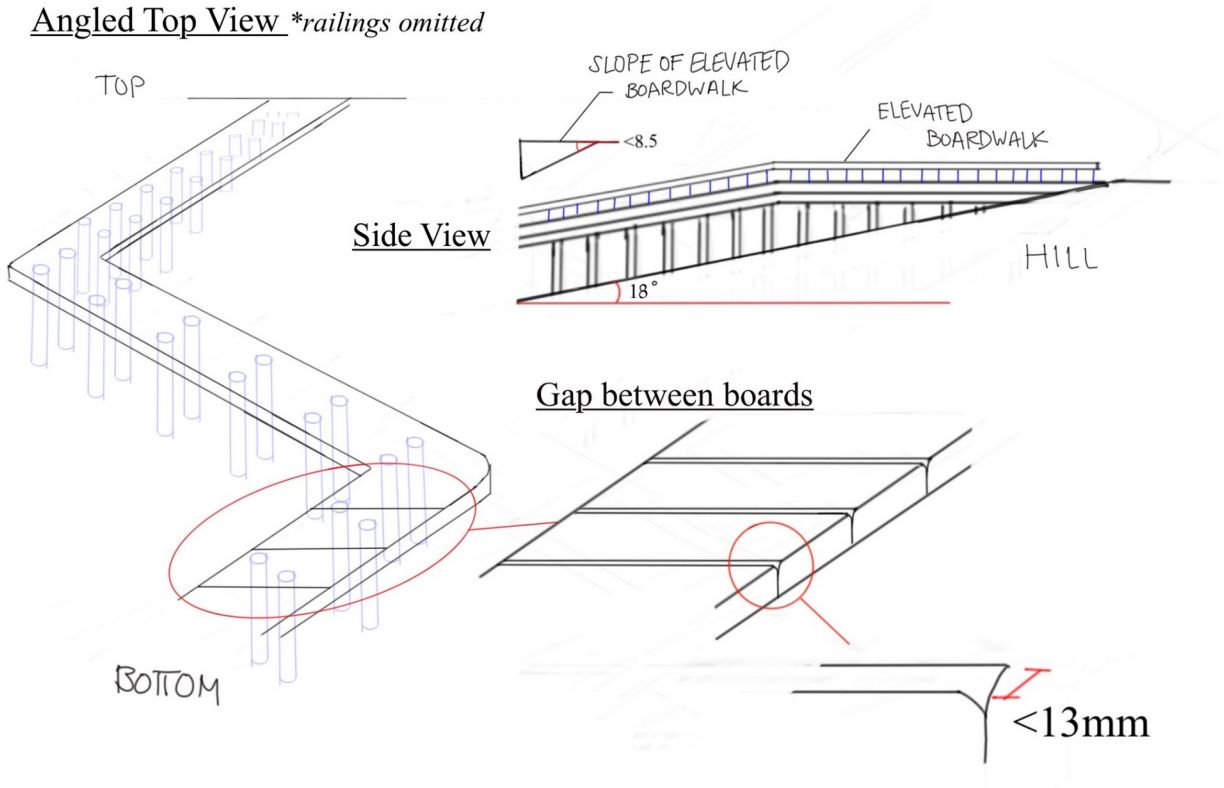



Figure 14: Top View and Side View of the Elevated Boardwalk

The Z-Shape of the Elevated Boardwalk overcomes the steepness of the slope by introducing an easier to navigate gradual decline (Figure 14). This allows for easy movement up and down the hill. The elevated platform conserves the environment as the area of the Black Oak Savannah will only be removed by the placement of poles to support the boardwalk. This allows for a wider area of occupied space to provide more free movement of wheeled aids. Ultimately, it is more conservative of the environment, preventing the destruction of the Savannah while still being an accessible design. It stands out from the other solutions due to its minimal effects on the environment. It requires the least minimal of occupation of Savannah while requiring no energy to operate.

## 8.0 Conclusion

The High Park Project Requirements document the inequitable access in the High Park area between the Grenadier Café and Grenadier Pond. The site visit indicated that some pathways needed revision. Various tripping hazards and steep slopes were observed within the paved pathways and the accessible routes were more than triple their walking distance.

As such, the design solution will focus on transporting people with mobility issues from the café to the pond and maintain the Black Oak Savannah. The design objectives emphasize minimizing the impact on the land, travel time, cost, and energy efficiency. Most importantly, it must abide by the laws and standards associated with accessibility and natural conservation.

The document provides the necessary guidelines when considering the accessible design solutions to optimize and protect the nature experience at High Park.

Following a six-step idea generation process and a four-step idea selection process, solutions were narrowed down to three alternative designs: The Elevated Boardwalk, Birdseye Gondola; and the Graded Switchback. Weighing their fulfilment of the functions and objectives, the group proposes the Elevated Boardwalk as the solution.

The boardwalk outperforms the other two designs in the most significant categories and accomplishes the client’s request by providing a safe, community-driven, and accessible method of transportation that conserves the Black Oak Savannah. To proceed with the project, the client must choose which of the alternative designs to implement. After approval, construction can commence.

## 9.0 Appendix

### Appendix A: AI Usage

General Use:

1. ChatGPT: Used to find sources to justify objectives and their metrics, provide examples of solutions to boost the process of idea generation and reorder reference list
2. Copilot: Used to create citations for sources which were then edited and reviewed

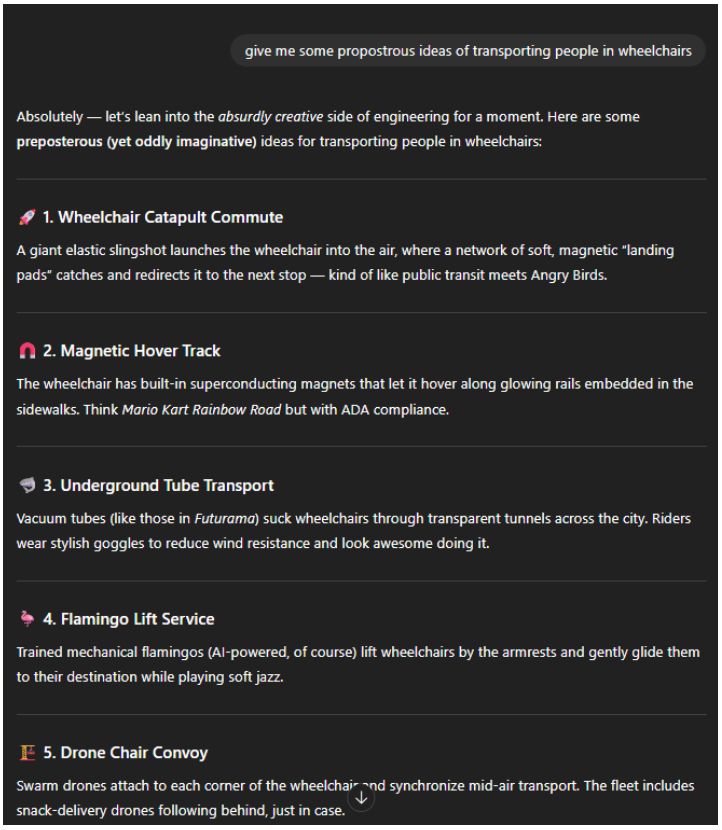
Specific Use:

1. ChatGPT

Prompts Used:

- "Give me some preposterous, logical, and creative ideas of transporting people in wheelchairs"
- "Give me basic underground solutions and aerial supported systems"
- "Restart the process and think of super creative kinetic systems"

To integrate ChatGPT prompts into the design process, our group used creative AI generation not to specifically give design ideas but to spark group brainstorming topics from which our group can use idea generation methods such as SCAMPER to elaborate on human developed ideas.



think of something that is logical but super creative

Perfect — you want that sweet spot between *plausible engineering* and *wild creativity*. Here are some **logical but super imaginative** ideas for transporting people in wheelchairs that could *actually* inspire real designs:

### ⚙️ 1. Modular Track Pods

Imagine magnetic "smart tiles" laid into sidewalks or building floors.
Wheelchairs can **lock onto the track**, and the floor itself provides propulsion — like a conveyor belt you can join or leave anywhere.

- Can auto-adjust speed and direction
- Integrates charging pads for powered chairs
- Looks like futuristic moving walkways

### 🌿 2. The "Green Glide" Pathway

A network of **slightly elevated glass pathways** runs through parks and campuses. Beneath the surface: tiny electric rollers that gently move wheelchairs forward when the user taps a handle control.

- Feels effortless
- Powered by solar strips embedded in the glass
- Can light up at night for navigation

### 🚁 3. Drone Dock & Lift System

Instead of lifting the entire wheelchair, **a modular drone dock** (like a mini hover platform) locks underneath it.

- Wheelchairs drive onto a landing pad
- Drones latch magnetically and lift it just a few meters off the ground
- Can glide to another pad autonomously
  *Think "Uber for the sky," but safe and close to the ground.*

give me some more basic underground solutgions

Great direction — going **underground** keeps things weather-proof, discreet, and potentially energy-efficient. Here are some **basic but creative underground transport concepts** for wheelchair users — ideas that are more practical than sci-fi, but still imaginative:

### 🚇 1. Mini Subway Tunnel System

A small-scale version of a subway — just for wheelchairs or mobility pods.

- Narrow tunnels (about 1 meter wide) run between key locations (buildings, bus stops, etc.).
- People roll into a small **platform elevator**, which lowers them into the tunnel.
- Their wheelchair locks into a small **motorized capsule** that drives through automatically.
- Could use magnetic or rubber-tire propulsion.
  - ✅ Smooth, weather-proof, quiet
  - ❌ High installation cost — best for large campuses or hospitals

### 🌀 2. Pneumatic Tube Transit

Like the mail tubes in old buildings, but human-sized and much gentler.

- Wheelchairs roll into a sealed capsule.
- Air pressure pushes the capsule through a smooth underground tube.
- Cushioned stops and airflow control make it feel like gliding.
  - ✅ Fast and efficient for short distances
  - ❌ Pressure control and safety engineering are tricky

### 🛗 3. Automated Conveyor Tunnel

Think of an underground **horizontal elevator**:

- Wheelchairs roll onto a low-speed conveyor track that runs under the floor.
- The track can move groups of chairs slowly betw buildings or across campus.
  - ✅ Cheap, modular, easy to maintain

how about arial supported systems

Excellent — **aerial-supported systems** are a fantastic direction if you want something that's both futuristic *and* feasible. You can think of them as **"wheelchair skyways"** — using the space *above* walkways instead of below them.

Here are some creative but logical aerial transport concepts for people in wheelchairs:

### 🚡 1. Cable-Guided Chair Pods (Mini Gondolas)

Think of **small personal gondolas** designed for wheelchair access.

- The wheelchair rolls directly into a pod at ground level.
- The pod locks magnetically to the floor, then lifts up and connects to a **suspended cable line**.
- Moves between towers or stations automatically.
  - ✅ Smooth, quiet, already proven tech (based on ski lifts)
  - ✅ Easy to electrify or solar-power
  - ❌ Needs good safety systems for weather and wind

**Creative twist:** transparent pods with built-in audio guidance and adjustable lighting for sensory comfort.

### 🛤️ 2. Overhead Rail System (Sky Track)

A slim **monorail-style track** runs overhead, carrying small carriers for wheelchairs.

- Each carrier has an arm that grips the rail from above and gently lifts the chair a few centimeters off the ground.
- The system can follow sidewalks or weave through indoor spaces.
  - ✅ Less intrusive than a full lift
  - ✅ Possible to retrofit above existing walkways
  - ❌ Needs consistent height clearance

**Bonus idea:** dynamic height control — the rail lowers when it detects a rider, then lifts back up to travel.

restart and think of super creative ideas

Perfect — let's scrap convention and go full-on **creative engineering mode** — ideas that make people say *"Wait... that could actually work?"*

Here are some **wildly imaginative but clever** ways to transport people in wheelchairs — focusing on originality, spectacle, and functional logic.

---

### 1. The "Sky Ribbon"

A soft, flexible, *ribbon-like magnetic path* suspended in the air, twisting through buildings and trees.

- Wheelchairs latch onto it magnetically.
- The ribbon gently flexes, letting riders **"surf the skyline"** at slow speeds.
- Solar panels line its upper surface; at night, it glows like a floating highway of light.

**Inspiration:** blend of maglev + suspension bridge + kinetic sculpture.

---

### 2. The Aerial Conveyor Serpent

A segmented, snake-like transport tube that *moves itself* through the air using undulating robotic motion.

- It has openings along its body where wheelchairs roll in.
- The "serpent" travels in a continuous loop around a park or campus.
- The segments flex to change direction — alive but mechanical.

**Visual:** imagine a metallic dragon gliding silently above ground, offering rides through its body.

---

### 3. Magnetic Cloud Loops

A series of **floating magnetic loops** arranged like stepping stones in the sky.

- Each loop creates a stable magnetic "bubble" for wheelchairs to sit in.
- Riders jump (magnetically) from loop to loop — each transfer assisted by small AI-guided drones.

**Bonus:** looks like floating halos across the skylin

how about kenetic systems

Now *that's* an exciting angle — **kinetic systems** open up a whole world of movement powered by motion, energy transfer, and interaction rather than engines or motors.

Let's think in terms of **systems that move people in wheelchairs using motion, balance, energy recycling, and clever physics** — not just machines.

Here are some **super creative kinetic transport concepts** that are still physically logical:

---

### 1. The Ripple Path

Imagine a pathway that behaves like the surface of water — when a wheelchair rolls onto it, the tiles **flex slightly**, creating a *ripple wave* that transfers energy forward.

- The wave itself helps push the next section of path up, creating continuous, self-propagating motion.
- Each tile has small hydraulic pistons that store and release the pressure energy.
  - ✅ Part mechanical, part organic — the user's own motion keeps the system alive.
  - 💡 Could work beautifully indoors or in parks.

---

### 2. Kinetic Energy Floor

Every roll of a wheel compresses special tiles that **harvest and store kinetic energy** (like the Pavegen system).

- The stored energy powers micro-motors that *assist* in propelling the next wheelchair user.
- Essentially: users charge the system by moving — creating a community-powered transit path.
  - ✅ Encourages sustainability and shared energy use.
  - 💡 Great for campuses or malls.

---

### 3. Weighted Pendulum Rail

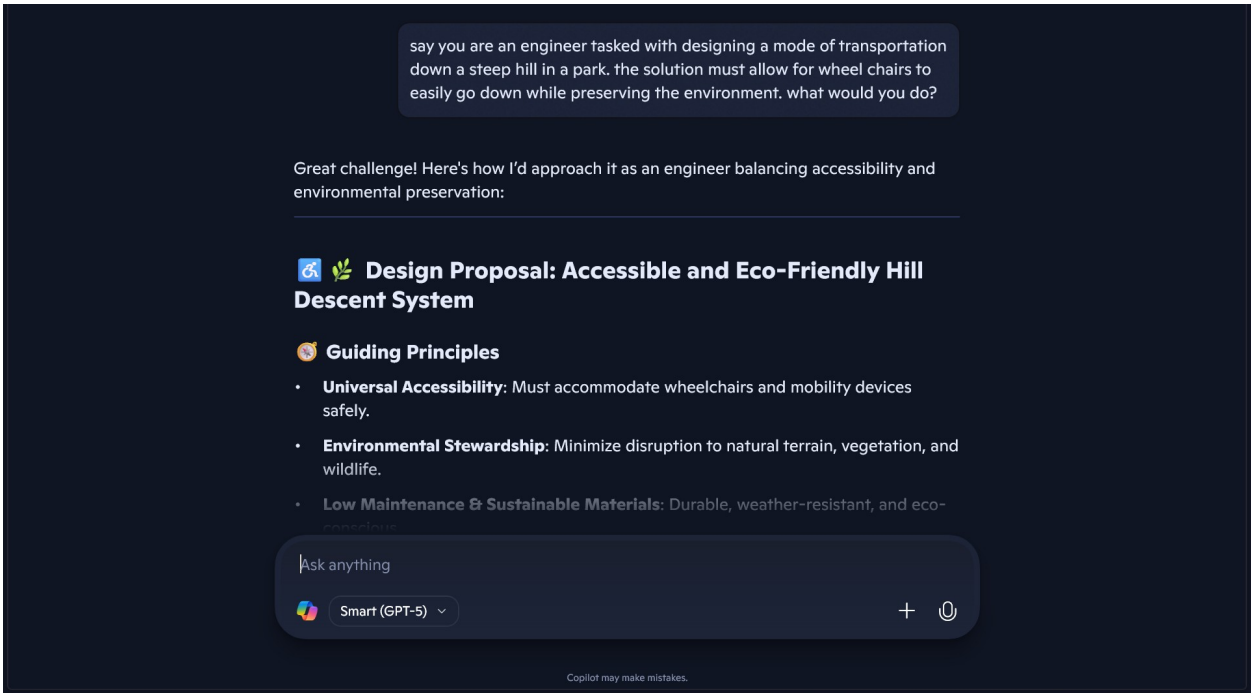
Aerial or underground track sections that move usin **unterbalanced pendulums**.

- When one section goes down, another lifts up.

2. Copilot

Prompts Used:

- Say you are an engineer tasked with designing a mode of transportation down a steep hill in a park. The solution must allow for wheel chairs to easily go down while preserving the environment. What would you do?



- **Environmental Stewardship**: Minimize disruption to natural terrain, vegetation, and wildlife.
- **Low Maintenance & Sustainable Materials**: Durable, weather-resistant, and eco-conscious.

---

### 🚡 Option 1: Inclined Elevator or Funicular

- **Description**: A rail-based cabin system that travels up and down the slope.
- **Pros**:
  - Fully accessible for wheelchairs.
  - Minimal footprint if built along existing clearings.
  - Electrically powered with potential for solar integration.
- **Cons**:
  - Higher upfront cost.
  - Requires careful integration to avoid visual and ecological disruption.

---

Ask anything

Smart (GPT-5)

Copilot may make mistakes.

### 🏞️ Option 2: Graded Switchback Path with Boardwalk

- **Description**: A gently sloped, winding path using permeable pavers or recycled composite decking.
- **Pros**:
  - Naturalistic and blends into the landscape.
  - Allows self-propelled or assisted wheelchair descent.
  - Can include rest areas and educational signage.
- **Cons**:
  - Requires more space due to switchbacks.
  - May need erosion control measures.

---

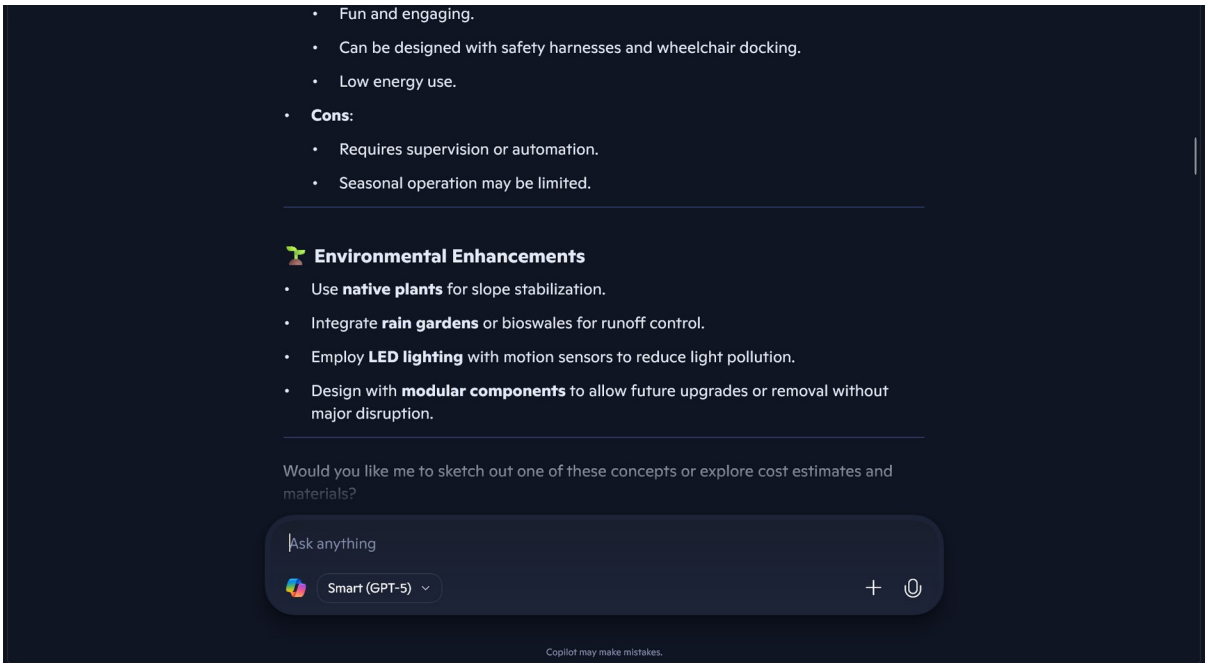
### 🛞 Option 3: Gravity-Assisted Track with Adaptive Carts

- **Description**: A guided track system where users board adaptive carts that descend slowly using gravity and braking systems.
- **Pros**:

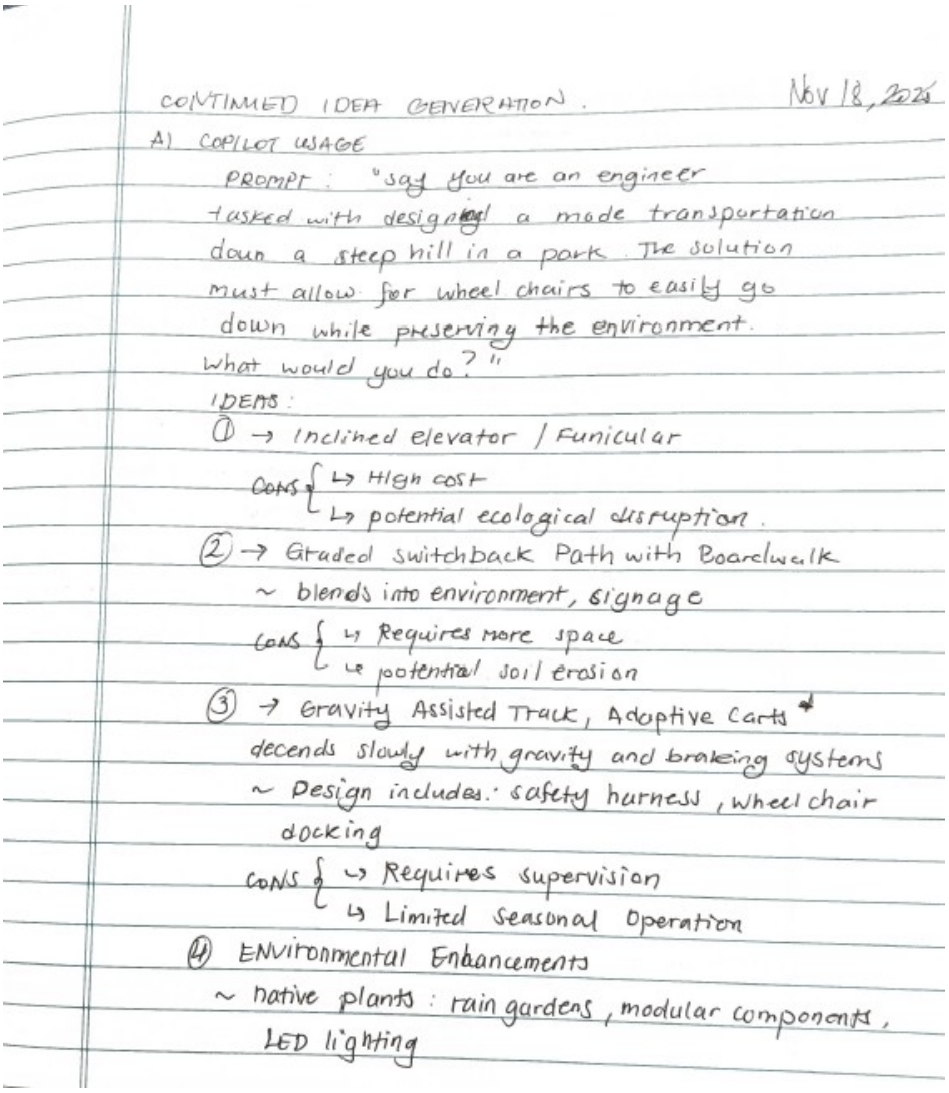
Ask anything

Smart (GPT-5)

Copilot may make mistakes.




List of Ideas generated (4):



CONTINUED IDEA GENERATION. Nov 18, 2025

A) COPILOT USAGE

PROMPT: "say you are an engineer tasked with designing a mode transportation down a steep hill in a park. The solution must allow for wheel chairs to easily go down while preserving the environment. What would you do?"

IDEAS:

① → Inclined elevator / Funicular

CONS { ↳ High cost
↳ potential ecological disruption.

② → Graded switchback Path with Boardwalk

~ blends into environment, signage

CONS { ↳ Requires more space
↳ potential soil erosion

③ → Gravity Assisted Track, Adoptive Carts

decends slowly with gravity and braking systems

~ Design includes: safety harness, wheel chair docking

CONS { ↳ Requires supervision
↳ Limited seasonal Operation

④ Environmental Enhancements

~ native plants: rain gardens, modular components, LED lighting

Process of Integration:

- To integrate these ideas into our list, we analyzed each one and talked about their feasibility which then led into modifications and additions to each solution to make them plausible in the context of the document

**Appendix B: Hazards on the Trail**

Images depicting the hazards in the existing trails from the Grenadier Café to the High Park pond area. These images were taken during our High Park site visit on October 4th between 2 PM to 4 PM. These photos show some of the service environment currently in the area.





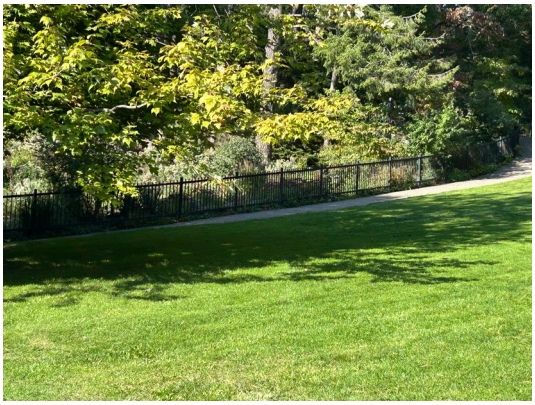
**Appendix C: Slope of Path**

Figure 4 depicts the steepness of the terrain in High Park. Canva was the tool used to highlight the various parts of the High Park terrain in Figure 4.



**Appendix D: Features of the Park**

This is an image of the handrails pedestrians were using to climb up the pathway discussed in the physical environment.





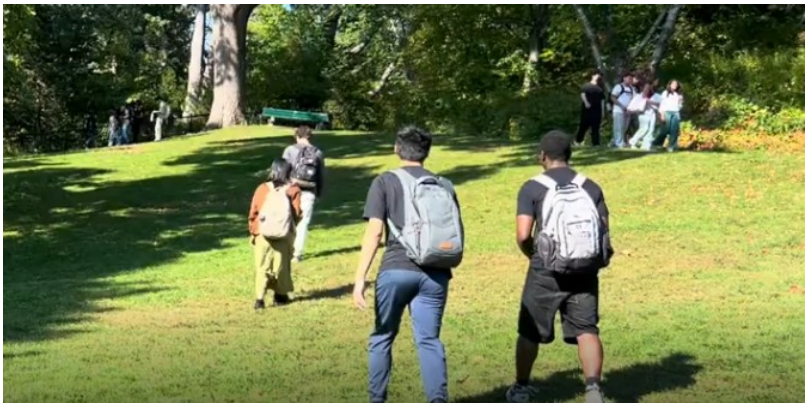

Pictured below are grass and trees in the Black Oak Savannah which is again a part of the physical environment.




In the Site Visit, there were a wide variety of people observed at the Park which is a part of the people seen as living things.




Grenadier Cafe
High Park
Cherry Blossoms
West Rd
Deer Pen Rd
Image taken
here
Portuguese Monument
Hillside Gardens
Hillside Rd



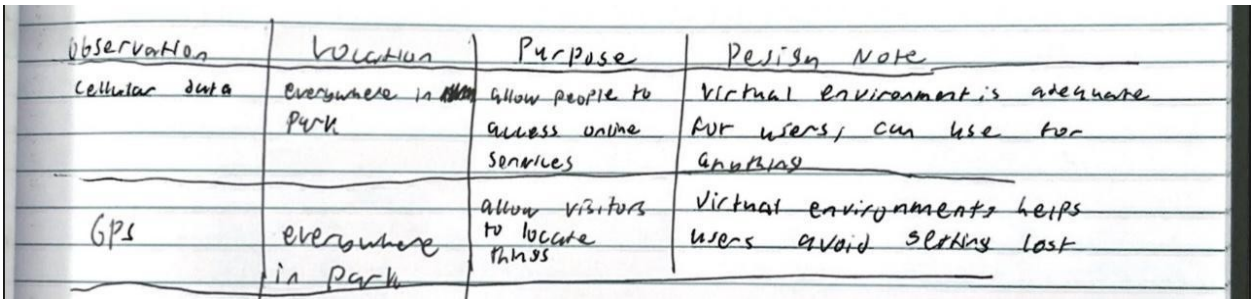
**Appendix E: Trackless Train**

The Trackless Train shown below transports many people around High Park as mentioned in the physical environment.



**Appendix F: Observation of Cellular Data and GPS in High Park**

During the High Park site visit, there was stable access to cellular data and GPS while in the area and the team used them for information gathering and location finding. The acknowledgement of the virtual environment was jotted down in the Engineering Notebook below.



| Observation | Location | Purpose | Design Note |
|---|---|---|---|
| cellular data | everywhere in Park | allow people to access online services | virtual environment is adequate for users, can use for analysis |
| GPS | everywhere in park | allow visitors to locate things | virtual environments helps users avoid getting lost |

**Appendix G: Animals in High Park**

There were many animals that naturally exist in High Park which are shown in the images below. All of these images were obtained from the High Park site visit. The remaining animals were noted in our observations in our engineering notebook.

Rabbit:



Chipmunk:



Ducks:



Bees:



Pigeon:



**Appendix H: Interest Holder Brainstorm**

This appendix contains the list of the brainstormed interest holders that weren't included in the final list for this document.

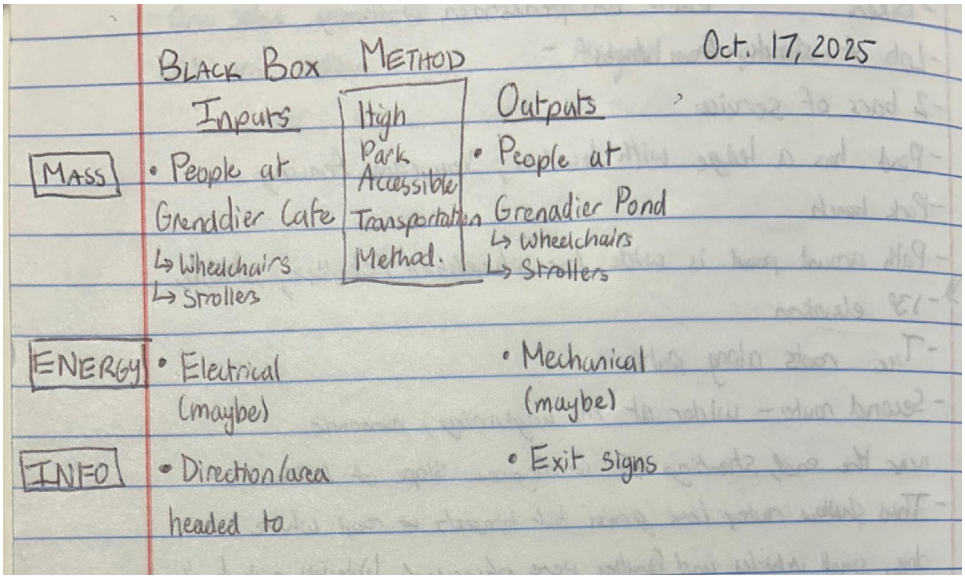
Table 9. List of Scrapped Interest Holders

| Interest Holders | Justification |
|---|---|
| High Park Access for All Group (Facebook group) [36] | The High Park Access for All Group concerns themselves with the accessibility of the park in terms of the High Park Movement Strategy [37]. |
| Destination Toronto (Marketing Organization) | This organization may want to advertise the design to appeal to interested viewers [38] [39]. |
| Turtle Protectors (Mishiikenh Gizhaasowin) | This group is responsible for protecting various species of turtles near the Grenadier pond and also in the Black Oak Savannah pathways [40]. |
| Canadian Association of Ergonomists | The Canadian Association of Ergonomists are concerned with the efficiency of the solution. They will evaluate the system to ensure it fits the needs of the people [41]. |
| Canadian Union of Public Employees | Employees managing the design will be provided employment and a fair wage [42]. |
| Grenadier Group (Managers of High-Park Café) [43] | The design will increase the accessibility to Grenadier Group's business towards those with accessibility issues, creating more revenue [43]. |
| Canadian National Institute for the Blind | The design will be accommodating for those with vision problems, referencing the safety standards promoted by CNIB [44]. |

**Appendix I: Extensive justification for Functional Requirements and Objectives**

Below are some of the methods and justifications used for determining the Functions and Objectives in sections 5.1 and 5.2.

The Black Box Method outlining inputs and outputs of the design to determine its functions.




The shortest distance from Grenadier Cafe to the pond.

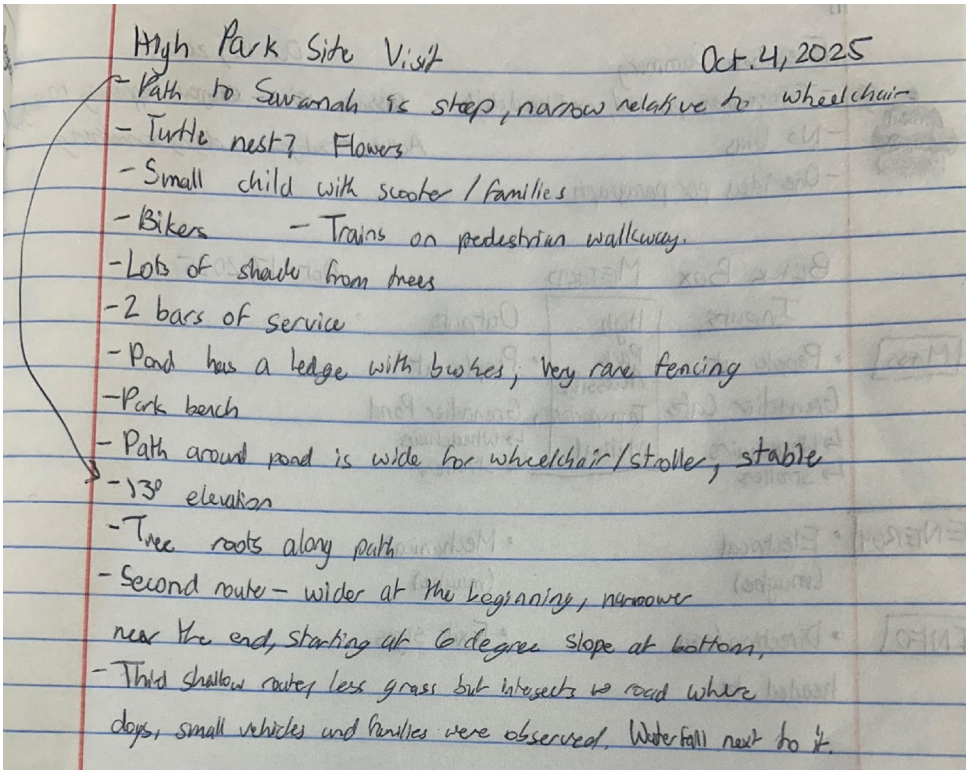



Objectives

- Energy Efficiency (An energy efficient system is accommodating to the environment around it as it is one of the main ways to reduce air pollution in an area. This is because energy produced in power plants can contain harmful air pollutants therefore minimizing the intake reduces pollution.)

**Appendix J: Site Visit Observations - Engineering Notebook**

The work pictured are excerpts from the team's Engineering Notebooks from the High Park site visit.



High Park Site Visit Oct. 4, 2025

- Path to Savannah is steep, narrow relative to wheelchair
- Turtle nest? Flowers
- Small child with scooter / families
- Bikers - Trains on pedestrian walkway.
- Lots of shade from trees
- 2 bars of service
- Pond has a ledge with bushes, very rare fencing
- Park bench
- Path around pond is wide for wheelchair/stroller, stable
- 13° elevation
- Tree roots along path
- Second route – wider at the beginning, narrower near the end, starting at 6 degree slope at bottom,
- Third shallow route, less grass but intersects to road where dogs, small vehicles and families were observed. Waterfall next to it.

OBSERVATION – HIGH PARK – GRENADIER POND Oct. 4, 2025

INCLINATION: ~~[illegible]~~° ~ 6°-20°, 14 och-

- hard to see, steeper near top, narrow pathways direction
- pathways near pond made of: dirt, stone
- lack of signage
- easy access to pond, not alot of safety signs.
  - ↳ not entirely blocked off
- dirt pathways through hill (café down to pond)
  - ↳ narrow, surrounded by shrubery, poking out

hillside garden hill near washrooms —

- people doing exercises, picnic
- open area
- people abiding by pathways (narrow)

~ pathway near fence pond. ~ 3.62 m wide

- train shares pathway with pedestrians
  - ↳ train supports elders, small children
  - ↳ bells to warn people
  - ↳ pedestrians move out away
- not many benches
- kids on scooters, many bikes
- minimal lighting

trash bins at beginning of route, not many throughout

- no railings along upward paths 13. Linda
- pathway through trees:
  - ↳ ~~narrow~~

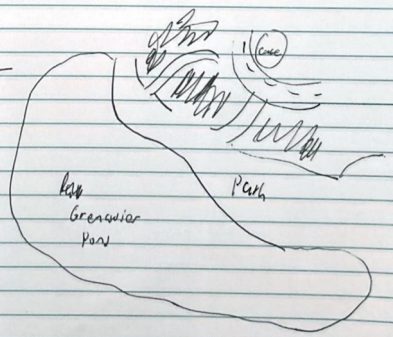
↳ lack of signage, vision obscured, minimal people :||

↳ adults without mobility impairment

High Park Site Visit: Oct. 4th 2025

- Site has many trees
- Large variety in paths (smooth, rugged, paved, gravelly, etc.)
- Squirrels
- entrance is accessible
- construction near the North side
- Lots of bikers, good community and many activities (tennis, statues and attractions)
- steep paths to lake ($20^0$–$15^0$)
- narrow pathy in savanna
- dominated by bushes
- train cars run along the path
- no railways for water (open access)
- Pond has many animals
- narrow and steep paths are very unaccessible with no guard rails
- lots of tourists near pond

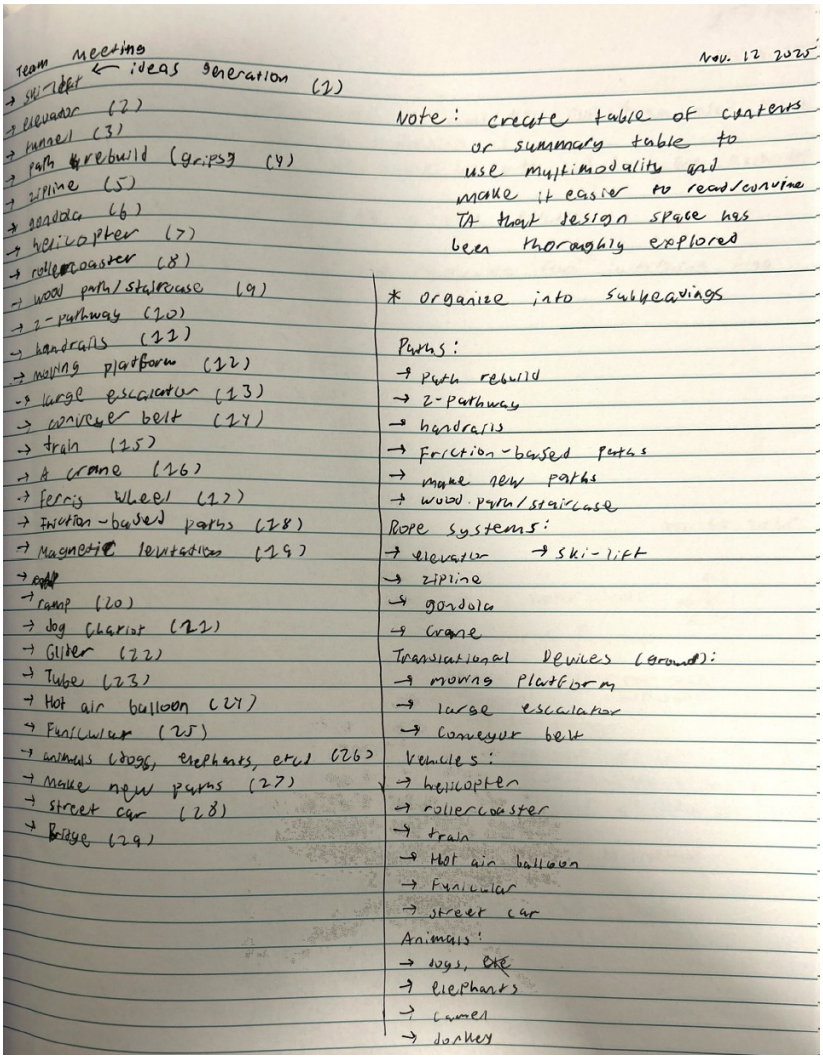
| Observation | Location | Purpose | Design Note |
|---|---|---|---|
| Cellular data | everywhere in Park | allow people to access online services | Virtual environment is alternate for users, can use for gaming |
| GPS | everywhere in park | allow visitors to locate things | Virtual environments helps users avoid getting lost |
| Unguarded water | at and near Grenadier Pond | | |
| handrails | Path going down to Pond | | |
| benches | along Path near Grenadier Pond | | |



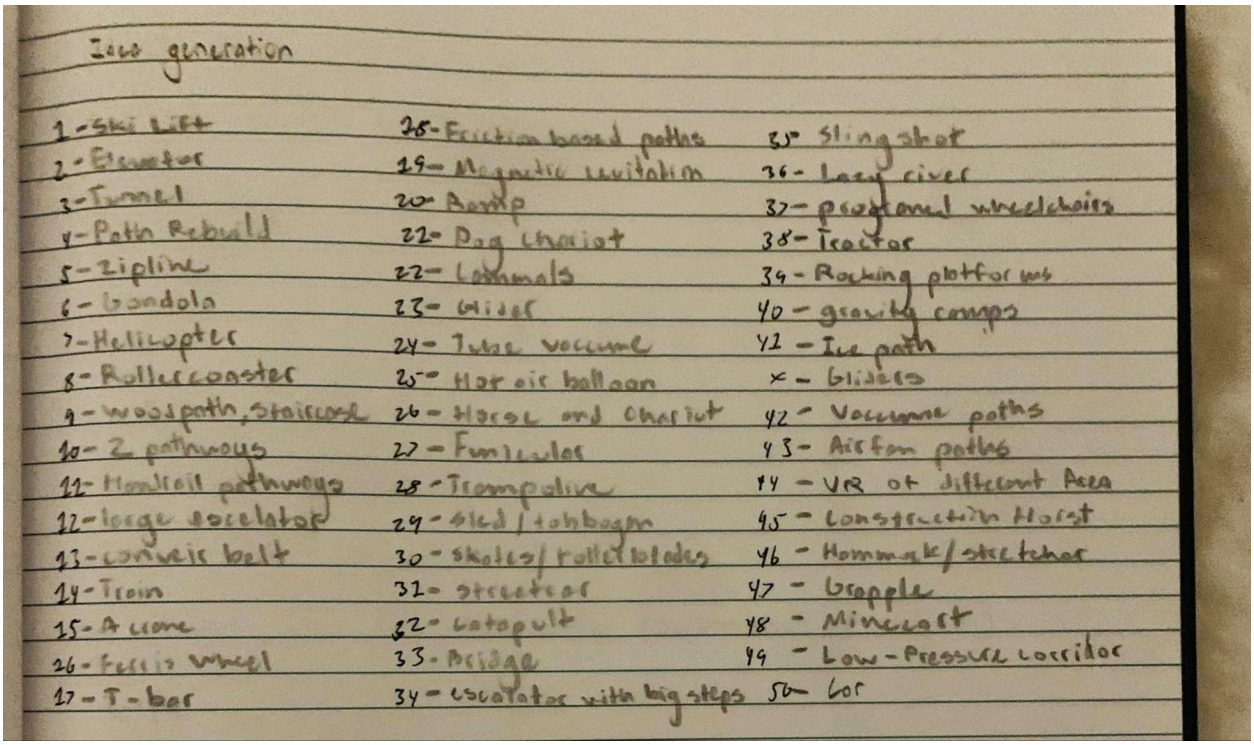
**Appendix K: Brainstormed Ideas List**

In our first team meeting, we generated 29 ideas that fit 5 subsections as shown below, covering changes to the paths, pulley systems, lateral movement devices, vehicles and animals.

Note: Idea one says ski lift



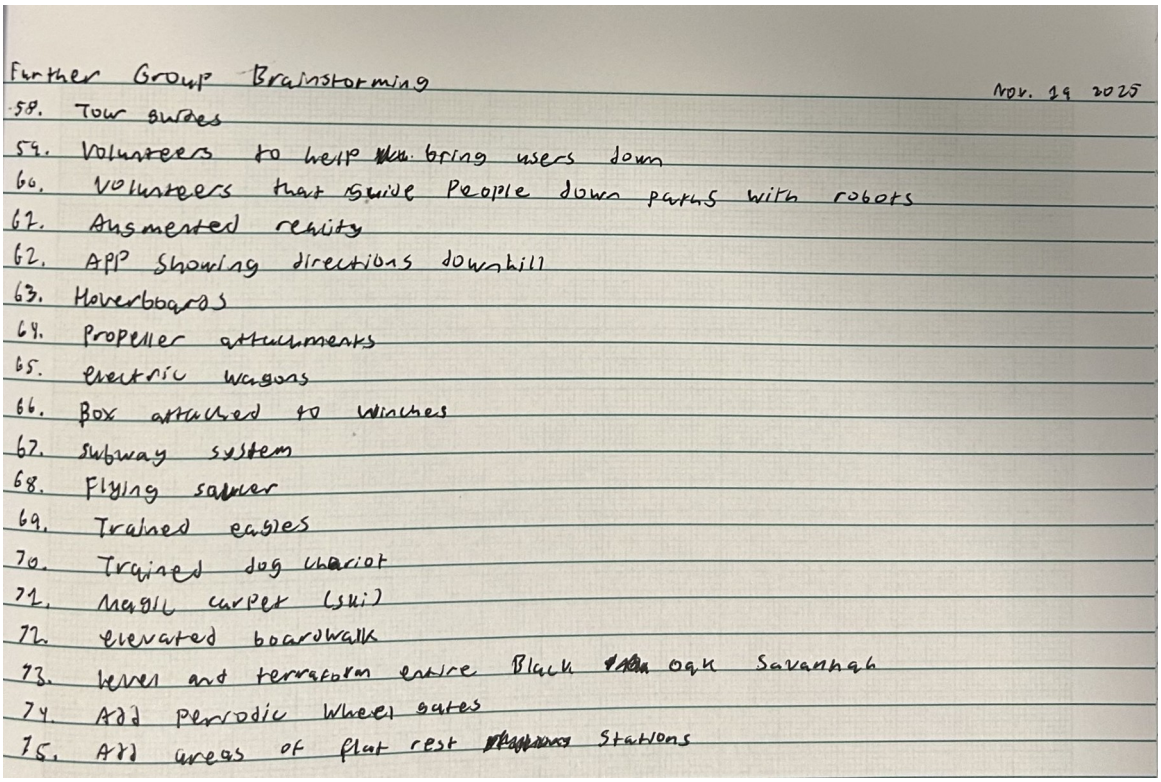

In our next brainstorming session, we generated a total of 50 ideas with the help of the morph chart which added more possible ideas to our current set of topics.



Idea generation

1 - Ski Lift
2 - Elevator
3 - Tunnel
4 - Path Rebuild
5 - Zipline
6 - Gondola
7 - Helicopter
8 - Rollercoaster
9 - Woodpath, staircase
10 - 2 pathways
11 - Handrail pathways
12 - Large escalator
13 - Conveir belt
14 - Train
15 - A crane
16 - Ferris wheel
17 - T-bar
18 - Friction based paths
19 - Magnetic levitation
20 - Ramp
22 - Dog chariot
22 - Carousels
23 - Glider
24 - Tube vacuum
25 - Hot air balloon
26 - Horse and chariot
27 - Funicular
28 - Trampoline
29 - Sled / toboggan
30 - Skates / rollerblades
32 - streetcar
32 - catapult
33 - Bridge
34 - escalator with big steps
35 - Sling shot
36 - Lazy river
37 - personal wheelchairs
38 - Tractor
39 - Rocking platforms
40 - gravity ramps
41 - Ice path
x - blisters
42 - Vacuume paths
43 - Airfan paths
44 - VR of different Area
45 - Construction Horse
46 - Hammock / stretcher
47 - Grapple
48 - Minecart
49 - Low-Pressure corridor
50 - Car

In our final group brainstorming session, we created 17 more solutions to add to our total list, finalizing our group brainstorming processes.

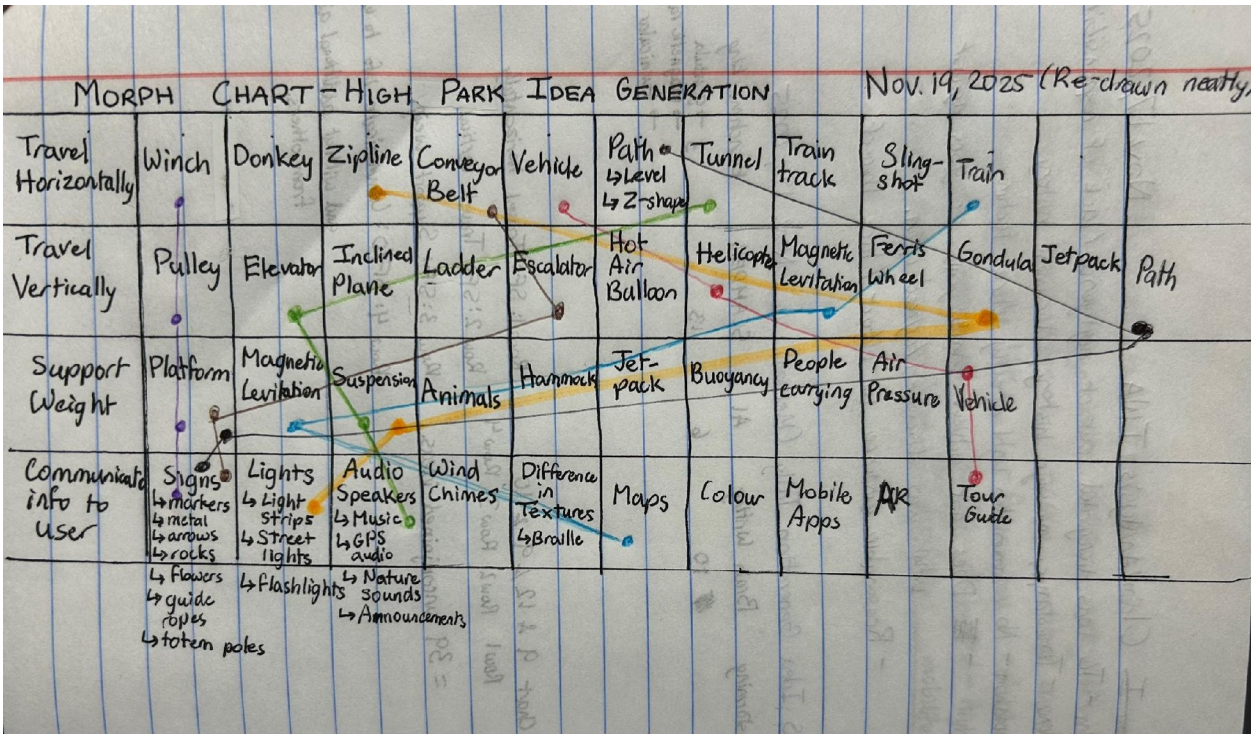


Further Group Brainstorming — Nov. 19 2025

58. Tour guides
59. Volunteers to help bring users down
60. Volunteers that guide People down paths with robots
61. Augmented reality
62. App showing directions downhill
63. Hoverboards
64. Propeller attachments
65. electric wagons
66. Box attached to winches
67. subway system
68. Flying saucer
69. Trained eagles
70. Trained dog chariot
71. Magic carpet (ski)
72. elevated boardwalk
73. levels and terraceform entire Black Oak Savannah
74. Add periodic Wheel gates
75. Add areas of flat rest Stations

**Appendix L: Morph Chart**

To generate more ideas, we created a morph chart on the board during the tutorial session and then further refined it in the engineering notebook which yielded some more useful ideas as seen below. The chart combines several pieces each meeting a specific detailed requirement to form a full design solution.

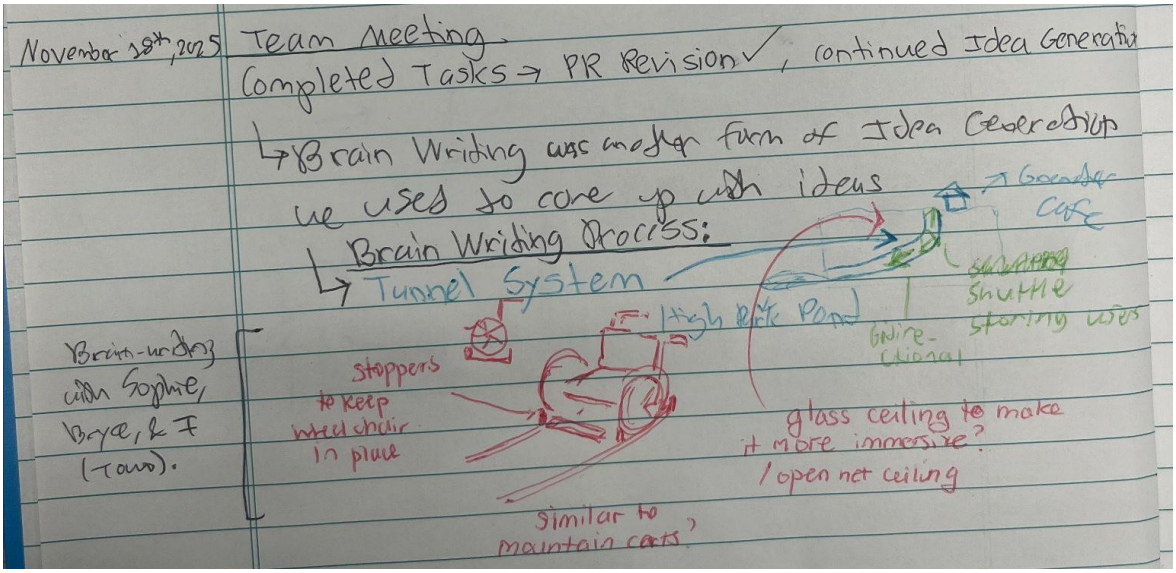



**Appendix M: Brainwriting Method**

The following images showcase the “Brain-Writing” idea generation method. This was done with group members Sophie (red), Bryce (green), and Towo (blue). Three separate ideas were generated using this method.

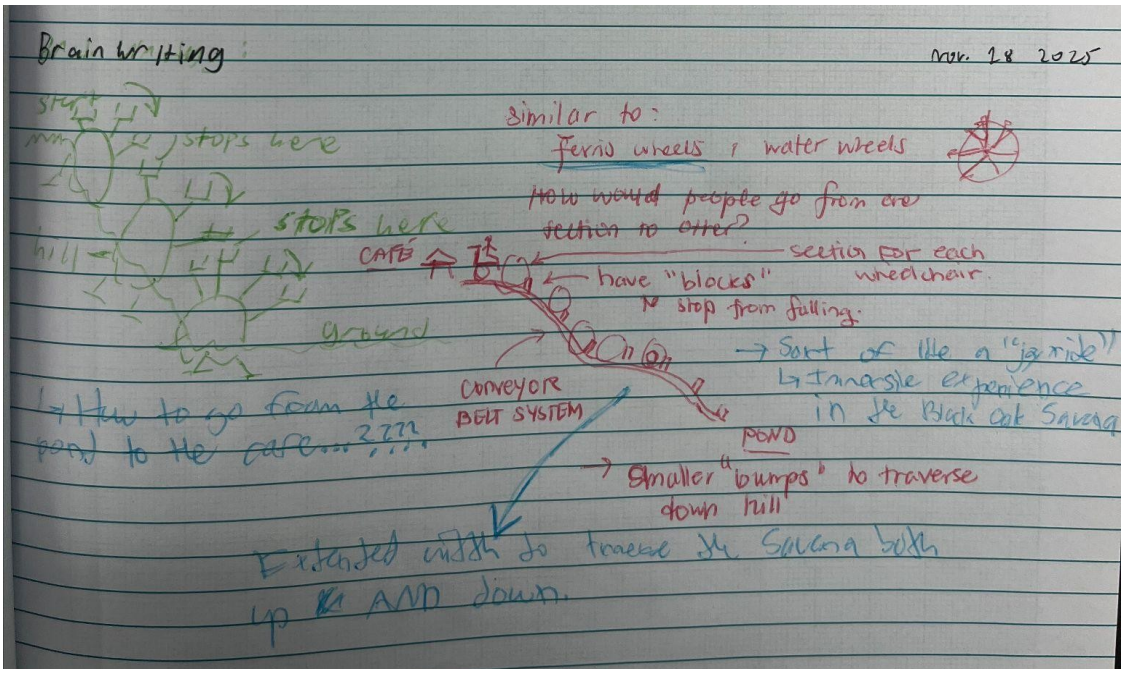
**Brain-Writing Idea #1:**

An underground tunnel system utilizing mini cars to transport people between Grenadier Cafe and Grenadier pond safely.



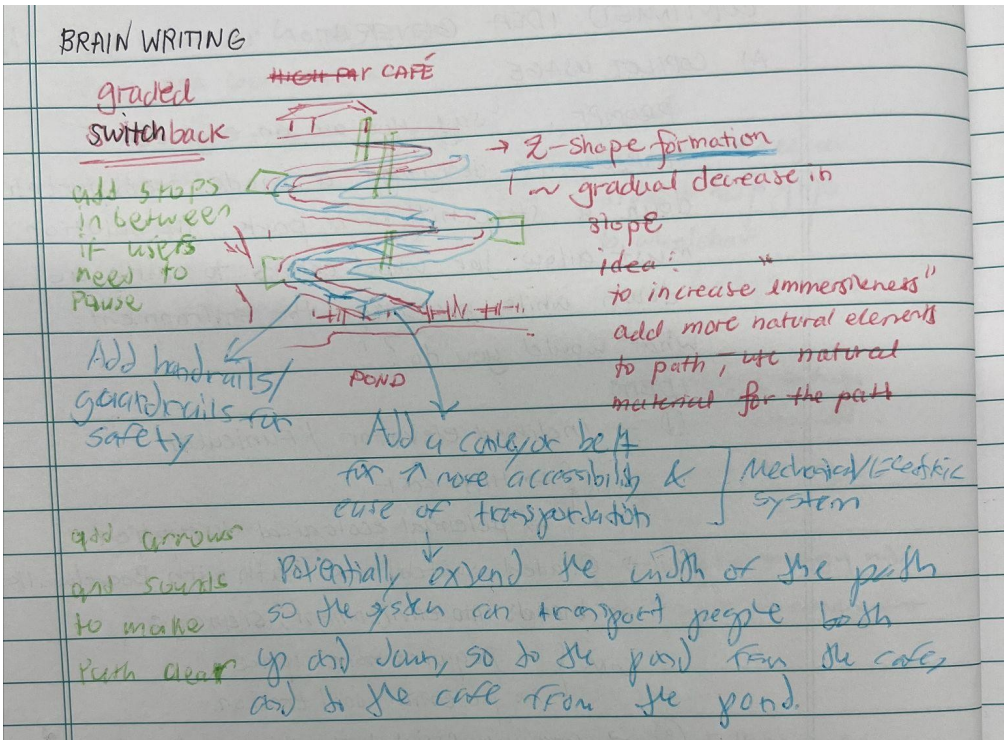

**Brain-Writing Idea #2:**

A water-wheel like system with a continuous set of wheels that move users down by rotating them to other platforms and allowing them to cross to lower wheels.




**Brain-Writing Idea #3:**

A graded switchback idea consisting of handrails and stops in between to allow users to gradually descend to the pond.



## Appendix N: Idea Generation Table

We compiled all of our ideas into a table and sorted them based on their means of functionality or their impact on the existing land. After numbering them all, we went through each idea and checked if they were duplicates by judging the similarities between their functions in the service environment.

Then, we went through each idea with the focus of keeping them within the constraints and ensuring that they follow the function. Most of our ideas passed this test after some modifications but some of them could not be fixed. These ideas usually violated the constraints of size, noise or slope but some of them also failed to transport a user with mobility issues effectively.

| Idea Type | # | Ideas | Duplicates (#) | Feasibility |
|---|---|---|---|---|
| Path Adjustments | 1 | Change paths to be made of sturdy wood and add signage | | YES |
| | 2 | Add handrails with arrows on ground | | YES |
| | 3 | Friction-based path with signage | | YES |
| | 4 | Add more natural elements and signage | | YES |
| | 5 | Gravity-assisted track with arrows | | YES |
| | 6 | Difference in texture to guide direction | | YES |
| | 7 | Add periodic wheel stoppers and arrows | | YES |
| | 8 | Add areas of flat rest stations with maps | | YES |
| | 9 | Level and terraform the entire Black Oak Savannah with signage | | NO |
| Introduce New Paths | 10 | Staircase with arrows | | NO |
| | 11 | Z-Shaped Pathways with Metal Signage | #11, #15 | YES |
| | 12 | Ramp with arrows on floor | | YES |
| | 13 | Elevated boardwalk with colored guide ropes | | YES |
| | 14 | Bridge with arrows | | YES |
| | 15 | Graded Switchback with arrows on ground | #11, #15 | YES |

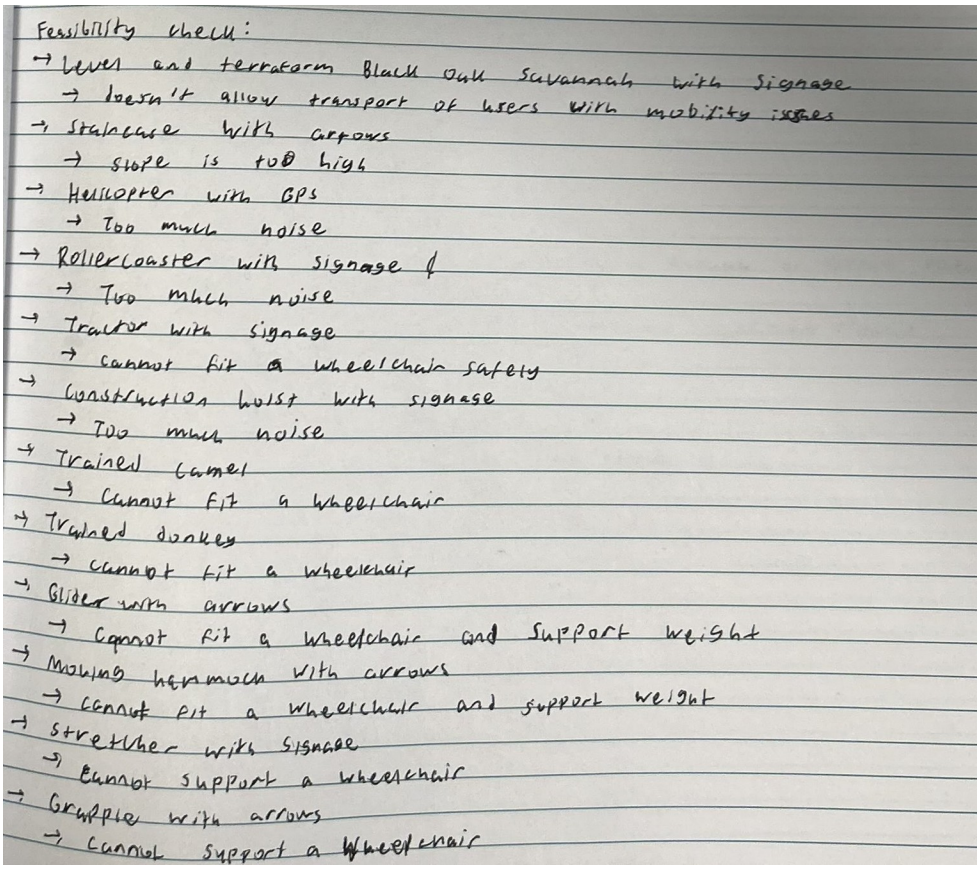
Table 10. Potential Design List Sorted by Functionality

| | | | | |
|---|---|---|---|---|
| | 16 | Magic carpet (ski) with arrows | | YES |
| Vehicle Travel | 17 | Helicopter with GPS | | NO |
| | 18 | Rollercoaster with Signage | | NO |
| | 19 | Train with Map | | YES |
| | 20 | Hot Air Balloon with Map | | YES |
| | 21 | Street Car with Maps | #21, #23 | YES |
| | 22 | Tractor with Signage | | NO |
| | 23 | Car with GPS | #21, #23 | YES |
| | 24 | Construction Hoist with Signage | | NO |
| Animal Travel | 25 | Trained Camel | | NO |
| | 26 | Trained Donkey | | NO |
| | 27 | Trained Dog Chariot | | YES |
| | 28 | Trained Horse Chariot | | YES |
| | 29 | Trained Eagles<br>Modify: With Ropes | | YES |
| Aerial Mechanism Movement | 30 | Ski Lift with Maps<br>Modify: Platformed Ski Lift | | YES |
| | 31 | Glider with arrows | | NO |
| | 32 | Zipline with arrows<br>Modify: Platformed Zipline | | YES |
| | 33 | Gondola Lift with Audio Speakers | | YES |
| | 34 | Crane with Audio Speakers | | NO |
| | 35 | Funicular with Audio Speakers | | YES |
| | 36 | Moving Platform with Arrows | #36, #37 | YES |
| | 37 | Rocking Platform with Arrows | #36, #37 | YES |
| | 38 | Moving Hammock with Arrows | | NO |
| | 39 | Stretcher with Signage | | NO |
| | 40 | Grapple with Arrows | | NO |
| | 41 | T-Bar with Arrows<br>Modify: Straps to user | | YES |
| | 42 | Flying Saucer with GPS | | YES |

| | | | | |
|---|---|---|---|---|
| Underground Travel | 43 | Tunnel with Arrows | | YES |
| | 44 | Tube Vacuum with Arrows | | YES |
| | 45 | Minecart system with Signage | | YES |
| | 46 | Subway system with Maps | | NO |
| Cable System Movement | 47 | Elevator with maps posted | | YES |
| | 48 | Box attached to winches with signage | | YES |
| Rotation Movement | 49 | Conveyor Belt with Arrows | #49, #51 | YES |
| | 50 | Ferris Wheel with Signage | | YES |
| | 51 | Platform over wheels with Arrows | #49, #51 | YES |
| | 52 | Continuous Water Wheel-esque System with Audio Speakers | | YES |
| | 53 | Large Escalator with Metal Signage | | YES |
| Suspension-based Movement | 54 | Magnetic Levitation<br>Modify: Magnetic Levitation Rail with Lights | | YES |
| | 55 | Trampoline with signage | | NO |
| Ground-based travel | 56 | Sled with arrowed path<br>Modify: With rope | | YES |
| | 57 | Toboggan with arrowed path<br>Modify: With rope | | YES |
| | 58 | Skates with arrowed path | | YES |
| | 59 | Specially Programmed Wheelchairs with GPS | | YES |
| | 60 | Hoverboards with arrows | | YES |
| | 61 | Electric wagons with arrows | | YES |
| Launch Based Mechanisms | 62 | Catapult with signage | | NO |
| | 63 | Slingshot with signage | | NO |
| Water-Based Movement Systems | 64 | Lazy River with arrows | | NO |
| | 65 | Ice Path with arrows | | NO |
| | 66 | Water slide with arrows | | NO |
| Air-Based | 67 | Propeller Attachment with signage | | YES |

| | | | | |
|---|---|---|---|---|
| Movement Systems | 68 | Air-Fan paths with arrows | | NO |
| | 69 | Low-Pressure corridor with signage | | YES |
| Virtual | 70 | App showing directions downhill | | NO |
| | 71 | Virtual Reality | | NO |
| | 72 | Augmented Reality | | NO |
| Human Aids | 73 | Individual tour guides | | YES |
| | 74 | Volunteers to help wheel down mobile aids | | YES |
| | 75 | Volunteers that stand at stations and serve as guides down path using robots | | YES |

When ruling out each idea, we wrote down reasons why they were unfeasible below.



→ subway system with maps
  → Too much noise
→ trampoline with signage
  → cannot transport wheelchairs horizontally enough or support them
→ catapult with signage
  → cannot support a wheelchair to achieve sufficient distance
→ Slingshot with signage
  → cannot support a wheelchair
→ Lazy river with arrows
  → doesn't provide transport between sites
→ Ice path with arrows
  → doesn't provide transport between sites
→ Water slides with arrows
  → doesn't provide transport between sites
→ Air fun path with arrows
  → doesn't support wheelchair
→ App showing directions downhill
  → doesn't provide transport between sites
→ Virtual reality
  → doesn't provide transport between sites
→ Augmented reality
  → doesn't provide transport between sites

**Appendix O: Multi-Voting Process**

To narrow down our ideas to about roughly ten ideas, the group had a vote to decide which ideas were best by each voting for our five favourite ideas as shown below. With five votes for each member, there were a total of 25 votes.

We chose five votes because we knew that there were 3-4 popular choices among every group member and the extra votes would let us include our personal favourites. This ensured that we could consider more out-of-the-box ideas. This was successful as our voting narrowed our list down to 11 ideas. The colors shown indicate the ideas that best meet the corresponding objective.

Table 11: Summary of our multi-voting process

| Name: | Idea #1 | Idea #2 | Idea #3 | Idea #4 | Idea #5 |
|---|---|---|---|---|---|
| Adrian | Gondola Lift with Audio Speakers | Handrail Pathways with Arrows on Ground | Z-Shaped Pathway with Metal Signage | Magnetic Levitation Rail with Lights | Elevators with Maps posted |
| Towo | Z-Shaped Pathway with Handrails and Metal Signage | Elevated Boardwalk with Colored Guide Ropes | Gondola Lift with Audio Speakers | Funicular with Audio Speakers | Elevators with Maps posted |
| Bryce | Contin- uous Water Wheel-esque System (Water as guide) | Funicular with Audio Speakers | Gondola Lift with Audio Speakers | Z-Shaped Path with Handrails and Metal Signage | Conveyor Belt and Lights |
| Lina | Elevated Boardwalk with Wooden Signage | Magnetic Levitation Rail with Lights | Elevator with Maps posted | Gondola Lift with Audio Speakers | Large Escalator with Metal Signage |
| Sophie | Z-Shaped Pathway with Metal Signage | Conveyor Belt and Lights | Elevated Boardwalk with Wooden Signage | Funicular with Audio Speakers | Friction Pathway with Metal Signage |

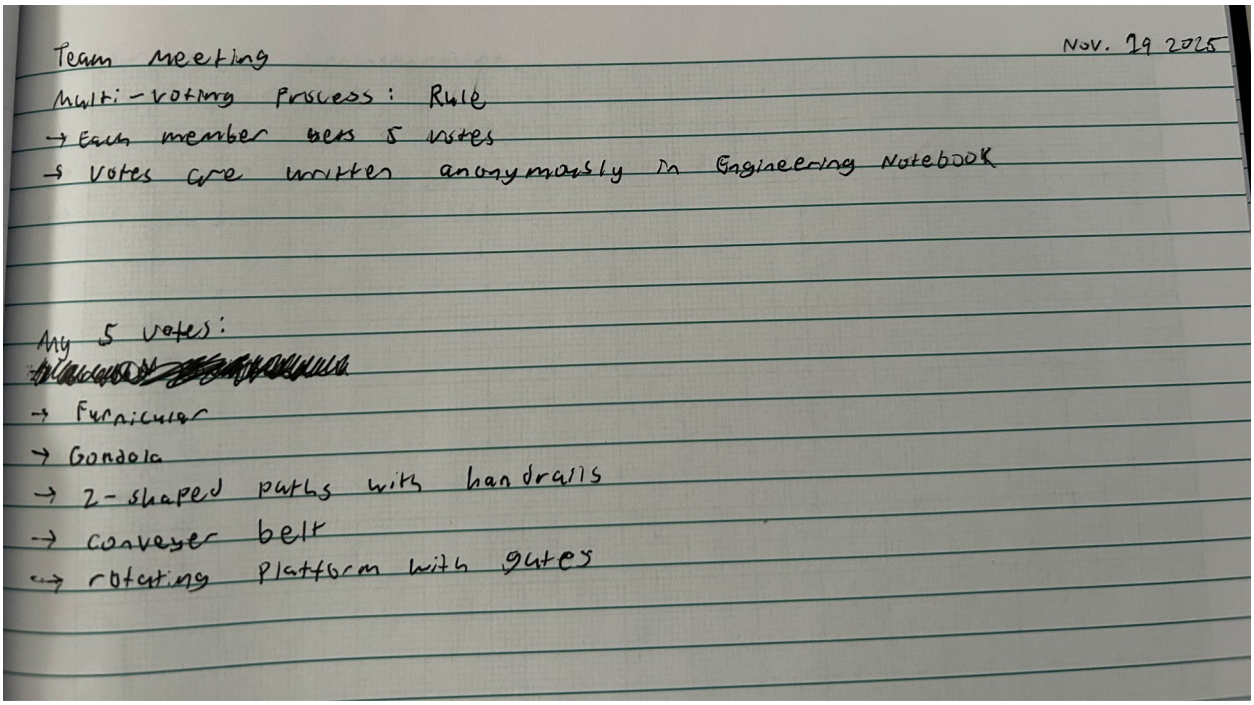
Objectives Legend:
= Maintaining the Black Oak Savannah

= Energy Efficient
= Transport Users without Additional Aid from Others

**Final List:**

1. Z-Shaped Pathway with Metal Signage
2. Magnetic Levitation Rail with Lights
3. Elevators with Maps posted
4. Funicular with Audio Speakers
5. Gondola with Audio Speakers
6. Large escalator with Metal Signage
7. Elevated boardwalk with Wooden Signage
8. Friction pathway with Metal Signage
9. Conveyor Belt and Lights
10. Continuous Water Wheel System (Water as guide)
11. Handrail Pathway and Arrows on Ground

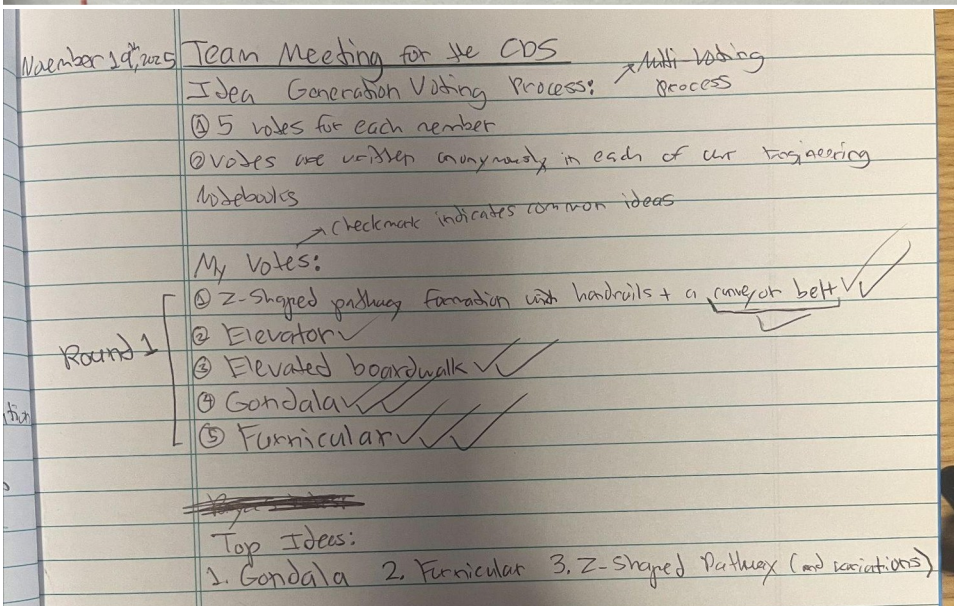
Below are excerpts from our engineering notebook from the multi-voting process



Multi-voting Process – Team 137 Nov. 19, 2025

Lina's Votes from Idea Generation

1. Elevated Boardwalk
2. Gondola/Funicular ✓✓✓
3. Large Escalator
4. Magnetic levitation ✓
5. Train



November 19th, 2025 Team Meeting for the CDS

Idea Generation Voting Process: ~~Multi-Voting~~ Process

① 5 votes for each member

② Votes are written anonymously in each of our Engineering Notebooks

My Votes: → checkmark indicates common ideas

Round 1

① Z-Shaped pathway formation with handrails + a conveyor belt ✓

② Elevator ✓

③ Elevated boardwalk ✓✓

④ Gondala ✓✓

⑤ Funnicular ✓✓

~~Top 5 Ideas~~

Top Ideas:

1. Gondala 2. Funnicular 3. Z-Shaped Pathway (and variations)

MULTIVOTING PROCESS — Nov 19, 2025

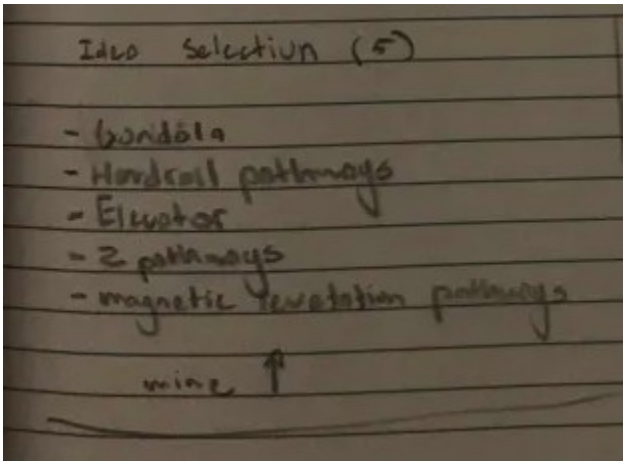
Rules:
- Each member has 5 votes
- Votes are written in engineering notebook
- Ideas with most votes are selected

If: cannot reach conclusion
↳ outliers are scrapped
↳ >1 votes are then voted on

CHOSEN Individual Ideas:
① Switch back ② Conveyor ③ Elevated Boardwalk
④ Funicular ⑤ Add materials that add friction

Team Ideas:

| | |
|---|---|
| Switchback: II | Z Path: I✗ |
| Gondola: III | Magnetic Levetation: II |
| Handrail: II | Conveyor: III |
| Elevator: III | Elevated Boardwalk: III |
| Funicular: IIII | Add friction: I |
| Escalator: I | Z shaped Conveyor: II |


Idea Selection (5)

- Gondola
- Handrail pathways
- Elevator
- Z pathways
- magnetic levetation pathways

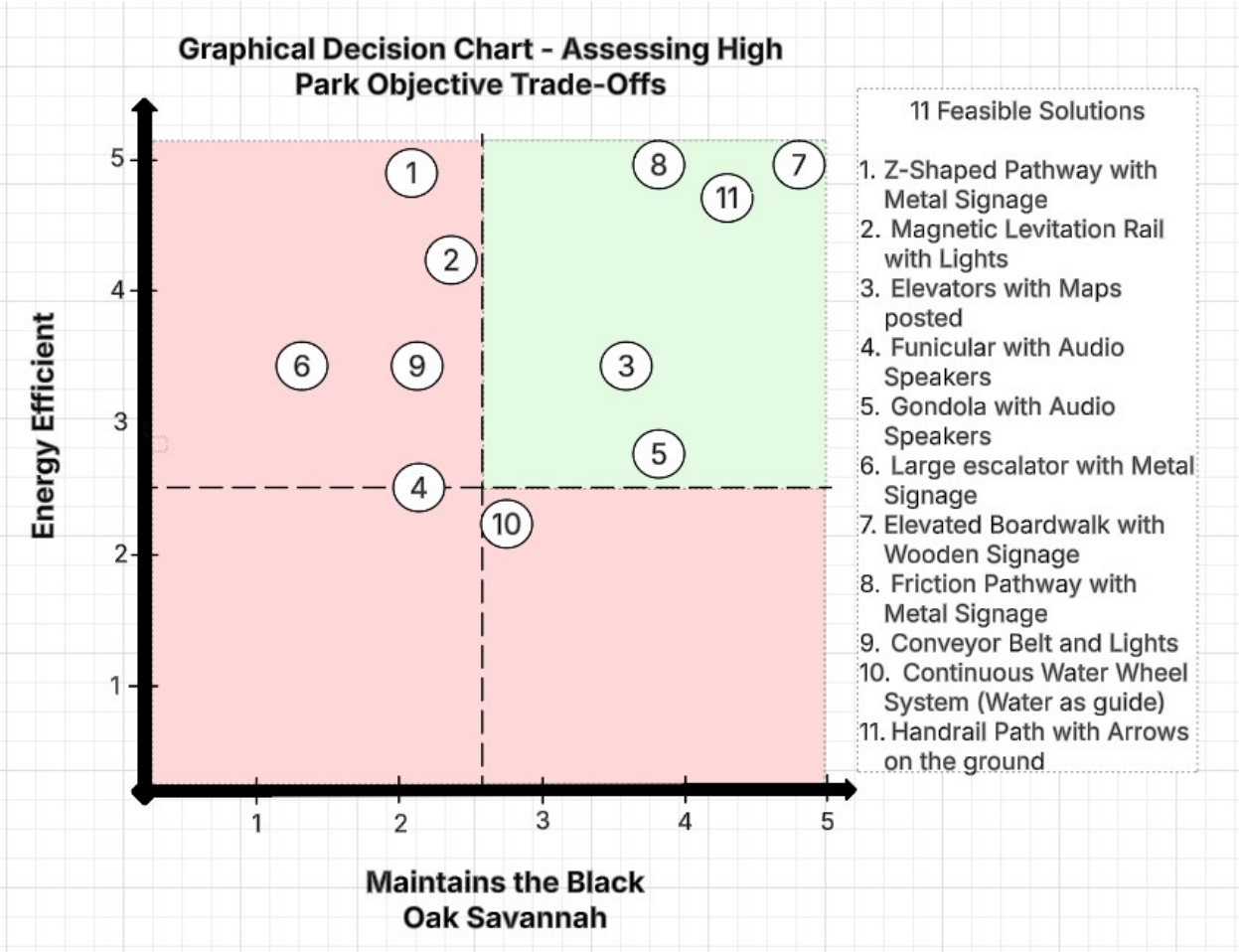
mine ↑

**Appendix P: Graphical Decision Chart**

The graphical decision chart was used to rank how well each of the voted solutions fulfill our outlined objectives. The two main objectives for the design in order of importance are maintaining the Black Oak Savannah and energy efficiency. See subsection 5.2 for details on how the designs were assessed. Based on the results of this chart and a group consensus using the FOCs as guidelines, three design alternatives were chosen.

We chose conservation of the Black Oak Savannah as our most important objective since it is an objective connected to our most significant interest holders and should be prioritized. Our second most important objective is energy efficiency since an energy intensive solution may be destructive to the environment and community.

The graph was taken from [45].



**Appendix Q: Research and justification for why the Gondola Lift is not as energy efficient as the Z-Shaped Elevated Boardwalk**

Gondola lifts can consume between 125 kW/h to 150 kW/h per system, and this number can increase based on the number of passengers on the gondola lift at a particular time [41].

**Appendix R: Detailed Description**

All (Lightweight) statements relate to the objective of increasing energy efficiency
All materials are found using [46].

Table 12. Elevated Boardwalk (Design 1)

| Components | Description |
|---|---|
| Materials | - Whole boardwalk will be made of Wood-Plastic Composite, (combination of recycled plastic and wood fibres), rot resistant (Low maintenance), Weather resistant due to polymer composition<br>- Signage made from polished oak wood, (low maintenance), environmentally friendly.<br>- Textured composite decking used as a nonslip surface to prevent incidents, low maintenance, environmentally friendly due to recycled plastic and wood origin. |
| Dimensions | - 4.00m wide,<br>- Length of 163.4m, length before each bend starting from the cafe (34.0m, 28.0m, 38.0m, 30.0m, 34.0m)<br>- Length between posts (2.50m)<br>- Land area removed (~9.64$m^2$)<br>- Radius of posts (0.1525m)<br>- Boardwalk maximum height (1.50m) |
| Lighting | - Only required for entrance and exit stations due to daytime operating hours (LED lighting) 4000 - 4500 natural light (good for environment) energy efficient, long lifespan (low maintenance), Brightness (100 lx) reduces stress on eyes + friendly to environment |
| Noise emission | - None |
| Safety | - Slope (18 degrees x (root(106^2 + 58^2)/163.4) = 13.31 degrees<br>- AED safety kits every 33 poles<br>- Smooth ramp entries for wheelchairs |
| General design | - 132 total support posts<br>- ~90 degree bends at each Z |

Table 13. Birdseye Gondola (Design 2)

| Components | Description |
|---|---|
| Materials | Cart<br>- Transparent gondola windows (Polycarbonate) lightweight, impact resistant, low maintenance |

| | |
|---|---|
| | - Metallic gondola frame (Aluminum alloy) lightweight, corrosion resistance (low maintenance)<br>- (Polyurethane) lightweight, thermal and noise insulation<br>- All internal components, rails, floor hangers, (Polished oak wood) environmentally friendly, low maintenance, low cost<br><br>Gondola Cable-line<br>- Cables (Galvanized Steel) galvanization reduces corrosion (low maintenance), low cost<br>- Towers (Standard Structural Steel) low corrosion, low cost<br>- Start and Finish areas (mechanical sections made of cast steel + stainless steel) reduces corrosion, (external sections made of oak wood and cement), environmentally friendly, low cost<br>- Pulleys (stainless steel), high cost, reduces corrosion, low risk due to material characteristics |
| Dimensions | Cart<br>- External: Length (2.00m) Width (1.50m) Height (2.00m)<br>- Internal: Length (1.80m) Width (1.30m) Height (1.80m)<br>- Radius of rails (2.00cm)<br>- Ramp Length (1.00m) extendable, slope of 5% for entry<br><br>Gondola Cable-line<br>- Cable length (400m rotates in a loop), Cable diameter (40.0 mm), Minimum bending radius of cable (1.20m)<br>- Tower height + cable height above ground level (6.00m), Tower radius (1.20m)<br>- Pulley radius (.150m)<br>- Total land removed ($3.39m^2$) |
| Lighting | - Only required for entrance and exit stations due to daytime operating hours (LED lighting) 4000 - 4500 natural light (good for environment) energy efficient, long lifespan (low maintenance), Brightness (100 lx) reduces stress on eyes + friendly to environment |
| Noise emission | Cart<br>- 40 dBA inside cart due to Polyurethane<br>- Audio Speakers to provide directional aid (max 50dBA)<br><br>Gondola Cable-line<br>- 50 dBA towards the environment due to Polyurethane applied around mechanical systems |
| Safety | Cart<br>- Contains AED for emergencies |

| | |
|---|---|
| | General system<br>- Mechanical emergency stop (manual)<br>- Automatic overspeed stop |
| General design | Cart<br>- Maximum shown interior weight (800 pounds)<br>- # of carts (4)<br><br>Gondola Cable-line<br>- 1 tower |

Table 14. Graded Switchback (Design 3)

| Components | Description |
|---|---|
| Materials | - Pathway (resin-bound gravel), high friction, environmentally friendly (no unnatural synthetic components)<br>- Signage made from polished oak wood, (low maintenance), environmentally friendly.<br>- Handrails made of Wood-Plastic Composite, (combination of recycled plastic and wood fibres), rot resistant (Low maintenance), Weather resistant due to polymer composition<br>- Arrows to show directions composed of acrylic paints, UV resistant, low maintenance, slightly expensive |
| Dimensions | - Pathway is 4.00m wide<br>- Length of 163.4m, length before each bend starting from the cafe (34.0m, 28.0m, 38.0m, 30.0m, 34.0m)<br>- Turning radius 1.5m<br>- Handrail height (1.00m)<br>- Radius of handrails (2.50cm) |
| Lighting | - Only required for entrance and exit stations due to daytime operating hours (LED lighting) 4000 - 4500 natural light (good for environment) energy efficient, long lifespan (low maintenance), Brightness (100 lx) reduces stress on eyes + friendly to environment |
| Noise emission | - None |
| Safety | - Slope (18 degrees x (root(106^2 + 58^2)/163.4) = 13.31 degrees<br>- AED safety kits every 41m (163.4/4) in case of emergencies |
| General design | - Drainage systems every 41m to reduce prevent flooding |

**Appendix S: Pugh Chart**

Table 15. Pugh Chart:

| | Alternative Designs | | | | |
|---|---|---|---|---|---|
| Objectives | Datum: Elevated Boardwalk | Gondola Lift with Audio Speakers | Justifications | Graded Switchback | Justifications |
| Maintain the Black Oak Savannah | S | +1 | Requires less of the savannah space to build and operate effectively since it can function above the savannah. | -2 | Requires the destruction of the savannah to create the Z-shape path which causes more destruction than the pillars of the boardwalk. |
| Energy Efficient | S | -2 | The Gondola Lift requires an extensive amount of energy to run, making it less energy efficient than the boardwalk design. [Appendix Q] | 0 | This solution and the elevated boardwalk both require no energy to function. |
| Transport Users without Additional Aid from Others | S | 0 | Audio speakers and wooden signage both provide equally adequate communication of info to users. | 0 | Both wooden signage and metal signage provide equally adequate communication of info to users. |
| Total | 0 | -1 | | -2 | |

The Z-Shaped Elevated Boardwalk was the standard used to compare the other two alternative conceptual designs as it's the "Datum" for the Pugh Chart. After using the Pugh Chart, the finalized design was the Elevated Boardwalk. Both the Birdseye Gondola and the Graded Switchback scored worse than the Elevated Boardwalk when it comes to meeting our objectives.

## Appendix T: Word Count Justification

To increase the detail and description of our alternative designs, we went over the objective goal for word count. We wanted to ensure that the client had all the necessary information and context connecting the solution and the detailed requirements to ensure that they could choose a solution with certainty. The higher word count provides more detail on how each design meets the functions and how our proposed design best meets the client's wishes. This trade-off benefits the reader greatly as the extra words provide information that may not be clear to the reader in terms of functionality and shows how the designs compare to each other in-depth.

## Appendix U: Attribution Table

| Section | Student Names | | | | |
|---|---|---|---|---|---|
| | Sophie | Lina | Towo | Adrian | Bryce |
| Executive Summary | WD, MR, ET, FP | WD, MR, ET | | | MR, FP |
| Introduction | FP | | ET | ET | WD, MR, ET, FP |
| Problem Statement | WD, MR, ET, RS1, FP, OR1 | RS1, WD, MR, OR1, ET, FP | RS6, ET | ET | ET, FP |
| Service Environment | WD, ET, FP | ET | WD, ET, MR, FP | WD, ET | WD, MR, ET, FP. RS4 |
| Interest Holders | FP | | ET, MR, RS5 | WD,MR, ET | WD, MR, ET, FP |
| Detailed Requirements | WD, ET, RS2, FP | WD, RS2, ET, FP | WD, ET, FP | WD,AI2,AI3,RS3, MR, ET | WD, MR, ET, FP |
| Idea Generation | AI4, OR4, WD, ET | AI4, WD, FP | WD, MR | WD, AI4, OR11, ET, FP | WD, MR, ET, FP, OR4 |
| Alternate Design Selection | WD, OR4, ET | WD, OR5, FP | WD, ET | OR5, OR11, ET, FP | WD, MR, ET, FP, OR4 |
| Alternate Conceptual Designs | OR6, WD, OR9, ET | WD, FP | RS7, MR, ET, WD, OR7, OR8, FP | OR9, ET, RS7, OR11, FP | WD, MR, ET, FP |
| Proposed Conceptual Design | WD, ET, OR6, OR9, OR12 | WD, ET, OR1, FP | WD, MR, ET, FP, OR10 | | FP |
| Conclusion | MR, FP, ET | WD, MR, ET, FP | | OR11 | ET, MR, FP |
| Appendix | WD, ET | OR2, WD, ET | WD, ET, FP | WD, ET, MR, OR5 | MR, ET, FP, OR1 |
| Reference List | ET | WD, ET | WD, MR, ET, AI1 | ET, MR, WD, OR3 | WD, MR, ET, OR3, FP, AI5 |

Fill in abbreviations for the tasks you completed for each section of the report using the abbreviations below. You do not have to fill in every cell in the table.

RS – Research (give details below)

WD – Wrote Draft

MR – Major Revision

ET – Edited

FP – Final Proofread of COMPLETE DOCUMENT verifying for flow and consistency

AI – Used Generative AI (give details below)

OR – Other (give details below)

**If you put RS** (research) please add a number identifier such as RS1, RS2, etc. Give the research question below. Note: you are not limited to two research questions, add the correct number of entries below to capture your team's work.

| | |
|---|---|
| RS1: | Accessibility standards, slope inclinations, High Park Map |
| RS2: | Minimum wheelchair path width, black box method, secondary functions vs primary |
| RS3: | Benefits of energy efficiency, incident rates in Ontario systems, work wages, functional hours |
| RS4: | What are the weather and precipitation ranges in the area? |
| RS5: | Researched the original indigenous caretakers of the land as well as their practices |
| RS6: | Researched a proper term for conservation and defined how we'll use conservation for the entire CDS |
| RS7: | Researched how much energy a Gondola Lift requires to operate |

Some examples include: conducting site visits, identifying and reviewing articles for information, meeting with a librarian for help, etc.

**If you put AI** (used Generative AI) please add a number identifier such as AI1, AI2, etc. Explain how you used Generative AI in the section or document. Note: you are not limited to two AI usages, add the correct number of entries below to capture your team's work.

| | |
|---|---|
| AI1: | Used AI to draft a reference list which was vetted and edited based on the IEEE formatting document |
| AI2: | Used AI to find sources that objectives could be referenced to ie. an energy demanding project in a different Toronto park |
| AI3 | Used AI to find and convert metrics ie. define and change kW/h to kJ/h |
| AI4 | Helped generate additional ideas |
| AI5 | Used to reorder reference list |

Some examples include: generating a draft for you to review, idea generation, editing a section you drafted, etc.

**If you put OR** (other) please add a number identifier such as OR1, OR2, etc. Explain what you did below. Note: you are not limited to 'other' types of work, add the correct number of entries below to capture your team's work.

| | |
|---|---|
| OR1: | Developing figures for maps and highlighting routes |
| OR2: | Organizing site visit images and categories of appendix |
| OR3: | Reorganizing References |
| OR4: | Created flow charts |
| OR5: | Created graphical design chart + detailed description chart |
| OR6: | Created final design drawings of alternative designs |
| OR7: | Added a figure to indicate the location of the towers needed for the Gondola Lift design |

| | |
|---|---|
| OR8: | Created the Pugh Chart |
| OR9: | Created draft of design |
| OR10: | Added the side-by-side view of the existing pathway and the pathway our proposed solution follows |
| OR11: | Reduced Word count |
| OR12: | Figure of map overview |

OR covers any work you did to develop the document that is not described by writing, editing, and/or researching. Some examples include: developing figures or diagrams to support multimodality, organizing references, meeting with your CI for guidance (out of tutorial), etc.

By typing your name below to sign, you verify that you have:

- Read the attribution table and agree that it accurately reflects your contribution to the associated document.
- Written the sections of the document attributed to you as WD and that they are either entirely original, or you have included appropriate acknowledgement of AI usage.
- Accurately cited and referenced any ideas or expressions of ideas taken from other sources according to the standard specified by this course.
- Read the University of Toronto Code of Behaviour on Academic Matters and understand the definition of academic offense includes (but is not limited to) all forms of plagiarism. Additionally, you understand that if you provide another student with any part of your own or your team's work, for whatever reason, and the student having received the work uses it for the purposes of committing an academic offence, then you are considered an equal party in the offence and will be subject to academic sanctions.

Student #1 Name Lina Habib

Student #2 Name Sophie Santiago

Student #3 Name Obatotowo Oyediran

Student #4 Name Adrian Kostrioukov

Student #5 Name Bryce Lin

Student #6 Name

Student #7 Name

## 10.0 References


[1] High Park Nature, “High Park Facts and Figures.” Accessed Oct. 18, 2025. [Online]. Available: https://highparknature.org/article/facts-and-figures-about-toronto-high-park/

[2] Engineering Strategies and Practices. (2025). *Equitable Access to Toronto’s High Park*. [PDF Document]. Available: https://q.utoronto.ca/courses/411111/files/38939170?module_item_id=7075067

[3] High Park Nature, “Maps and Directions – Toronto’s High Park.” Accessed Oct. 18, 2025. [Online]. Available: https://highparknature.org/article/facts-and-figures-about-toronto-high-park/

[4] Google, “Google Maps,” Google Maps. Accessed: Oct. 18, 2025. [Online]. Available: https://www.google.com/maps

[5] Accessibility Standards Canada, “6. Common accessibility measures – CAN-ASC-2.1 Outdoor Spaces (Draft).” Accessed: Oct. 18, 2025. [Online]. Available: https://accessible.canada.ca/creating-accessibility-standards/can-asc-21-outdoor-spaces-draft/6-common-accessibility-measures

[6] City of Toronto, “High Park.” Accessed Oct. 18, 2025. [Online]. Available: https://www.toronto.ca/explore-enjoy/parks-recreation/places-spaces/beaches-gardens-attractions/high-park/

[7] High Park Nature Centre, Rare Plants of High Park: A Guidebook. Toronto, ON: High Park Nature, Jul. 2021. Accessed: Nov. 23, 2025 [Online]. Available: https://highparknature.org/wp-content/uploads/2021/07/Rare-Plants-Guidebook-PDF.pdf.

[8] “High Park Village Weather,” Celinmeteo. Accessed Oct. 18, 2025. [Online]. Available: https://www.celinmeteo.com/wdisplay/wxindex.php

[9] C. Hoffstetter, “Contour Map Creator.” Accessed: Nov. 18, 2025. [Online]. Available: https://contourmapcreator.urgr8.ch/

[10] City of Toronto, *Mammals of Toronto*. (2012). Accessed: Nov. 22, 2025. [Online]. Available: https://www.toronto.ca/wp-content/uploads/2017/08/9718-Biodiversity-MammalsBook-Division-Planning-And-Development-part1.pdf

[11] City of Toronto, “Parks and Recreation.” Accessed Oct. 19, 2025. [Online]. Available: https://www.toronto.ca/city-government/accountability-operations-customer-service/city-administration/staff-directory-divisions-and-customer-service/parks-forestry-recreation/

[12] High Park Nature Centre, “Land Acknowledgement.” Accessed: Oct. 18, 2025. [Online]. Available: https://highparknaturecentre.com/land-acknowledgement/

[13] City of Toronto, “Urban Forest Management.” Accessed Oct. 19, 2025. [Online]. https://www.toronto.ca/services-payments/water-environment/trees/forest-management/urban-forest-management/

[14] High Park Nature, “High Park Natural Environment Committee.” Accessed Oct. 18, 2025. [Online]. Available: https://highparknature.org/get-involved-with-high-park-nature/volunteer-in-toronto-high-park/high-park-natural-environment-committee/

[15] Disability Alliance Canada, “Introducing Disability Alliance Canada.” Accessed: Oct. 18, 2025. [Online]. Available: https://disabilityalliancecanada.ca/2025/04/04/introducing-disability-alliance-canada

[16] City of Toronto, Commissioner..., “Extension of High Park Trackless Train Contract...” Mar. 28, 2000. Accessed Oct. 19, 2025. [Online]. Available: https://www.toronto.ca/legdocs/2000/agendas/council/cc/cc000509/edp5rpt/cl002.pdf

[17] High Park Nature, “About Our Stewardship Program.” Accessed Oct. 18, 2025. [Online]. Available: https://highparknature.org/get-involved-with-high-park-nature/volunteer-in-toronto-high-park/high-park-stewards/about-our-stewardship-program/

[18] Federation of Metro Tenants’ Associations, “Spotlight on Tenant Associations: High Park Residents Association.” Accessed Oct. 18, 2025. [Online]. Available: https://www.torontotenants.org/spotlight_on_tenant_associations_high_park_residents_association

[19] High Park Tenants’ Association. “High Park Tenants’ Association.” Accessed Oct. 18, 2025. [Online]. Available: https://www.hpta.ca/

[20] Statistics Canada, “Census Profile, 2021 — Parkdale—High Park.” Accessed: Oct. 18, 2025. [Online]. Available: https://www12.statcan.gc.ca/census-recensement/2021/dp-pd/prof/details/page.cfm?

Lang=E&SearchText=Parkdale%2D%2DHigh%20Park&DGUIDlist=2013A000435081&GENDERlist=1,2,3&STATISTIClist=1&HEADERlist=0

[21] City of Toronto, *Accessibility Design Guidelines* (Apr. 2004). https://www.toronto.ca/wp-content/uploads/2017/08/8fcf-accessibility_design_guidelines.pdf

[22] High Park Nature, "High Park's Rare Black Oak Savannah." Accessed Oct. 18, 2025. [Online]. Available: https://highparknature.org/article/high-parks-rare-black-oak-savannah/

[23] FIS, *Sustainability Guide for Ski Resorts*. 2023. Accessed Nov. 18, 2025. Available: https://assets.fis-ski.com/f/252177/x/93a1eb34fa/sustainability_skiguide_final.pdf

[24] U.S. EPA, "Energy Efficiency." ENERGY STAR. Accessed: Oct. 18, 2025. [Online]. Available: https://www.energystar.gov/about/how-energy-star-protects-environment/energy-efficiency

[25] Toronto Parks & Recreation, *Survey Summary*. Accessed: Nov. 18, 2025. https://www.toronto.ca/wp-content/uploads/2018/01/859f-survey-summary-parks-trails-wayfinding-strategy-phase-two.pdf

[26] Technical Standards and Safety Authority, "TSSA Public Safety Report Edition 2024," TSSA, Ontario, Canada, 2024. Accessed: Oct. 18, 2025. [Online]. Available: https://www.tssa.org/sites/default/files/2024-09/TSSA%20Public%20Safety%20Report%202024-Sep19%2C2024.pdf

[27] Accessibility for Ontarians with Disabilities Act, 2005, S.O. 2005, c. 11.

[28] Ontario Ministry of the Environment..., *Environmental Noise Guideline – NPC-300.* Accessed: Nov. 20, 2025. [Online]. https://www.ontario.ca/page/environmental-noise-guideline-stationary-and-transportation-sources-approval-and-planning

[29] "Flowchart Maker," LucidChart. Accessed Nov. 22, 2025. [Online]. Available: https://www.lucidchart.com/pages/examples/flowchart-maker

[30] Outdoor Structures Australia, *Boardwalk Design Guide*, v3.0. (2005). Accessed Nov. 22, 2025. [Online]. Available:

https://wilsontimbers.com/media/wysiwyg/Design_Guides/Boardwalk_Design_Guide_3__1.pdf

[31] "Boardwalk Design Tips...," PermaTrak. Accessed Nov. 22, 2025. [Online]. https://www.permatrak.com/news-events/boardwalk-design-tips-choosing-the-correct-pile-size-and-spacing#

[32] Government of Ontario, "Ontario Energy Snapshot." Accessed: Nov. 22, 2025. [Online]. Available: https://www.ontario.ca/page/ontario-energy-snapshot

[33] "Resin Bonded vs. Resin Bound Aggregate Surfacing," dream pave. Accessed: Nov. 23, 2025. [Online]. Available: https://dreampave.ca/resin-bonded-vs-resin-bound-aggregate-surfacing/?utm_source=chatgpt.com

[34] S. Tahmasseby, "Aerial ropeway system — feasibility study in Doha, Qatar," J. Unmanned Veh. Syst., vol. 9, no. 4, pp. 92-11, 2021, doi: 10.1139/JUVS-2020-0028

[35] "A guide to wheelchair ramps," Southwest New Brunswick Service Commission. Accessed: Nov. 23, 2025. [Online]. Available: https://snbsc.ca/application/files/9316/8780/1390/Wheelchair_Ramps.pdf

[36] High Park Access for All Group. Facebook. Accessed Oct. 19, 2025. [Online]. https://www.facebook.com/groups/highparkaccessforall/

[37] City of Toronto, "High Park Movement Strategy Implementation." Accessed Oct. 19, 2025. [Online]. https://haveyoursay.toronto.ca/pr-high-park-movement-strategy

[38] "High Park," Destination Toronto. Accessed Oct. 19, 2025. [Online]. https://www.destinationtoronto.com/neighbourhoods/high-park/

[39] "About Destination Toronto." Accessed Oct. 19, 2025. [Online]. https://www.destinationtoronto.com/about-us/

[40] Turtle Protectors, "Supported Parks." Accessed Oct. 18, 2025. [Online]. https://turtleprotectors.com/supported-parks

[41] Association of Canadian Ergonomics. "Home." Accessed Oct. 19, 2025. [Online]. https://www.ergonomicscanada.ca/

[42] City of Toronto, “Collective Agreements.” Accessed Oct. 19, 2025. [Online]. https://www.toronto.ca/city-government/.../collective-agreements/

[43] “Grenadier Cafe.” Accessed Oct. 18, 2025. [Online]. https://www.grenadiercafe.net/

[44] Canadian National Institute for the Blind, “The Future We Demand.” Accessed Oct. 18, 2025. [Online]. https://www.cnib.ca/en/future-we-demand

[45] “Virtual Whiteboard Where Ideas Ignite,” LucidSpark. Accessed Nov. 22, 2025. [Online]. https://lucid.co/lucidspark

[46] MatWeb, “MatWeb: The Online Materials Information Resource,” Accessed: 23-Nov-2025 [online]. Available: https://www.matweb.com/